

Certification, not simulation: the viability of harvest rules on two real resource systems, and the horizon of what can be certified

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Abstract

The general claim. Reference points and management rules are usually defended by *simulation*: a rule is run, the trajectory is inspected, and the rule is recommended. This paper defends them by *certification* instead, on two real resource systems, and argues that the difference is not stylistic. A simulation shows what happened on the runs that were tried; a certification states *why*, locates the verdict in a constant, and — most importantly — makes explicit *the horizon over which the verdict holds*. That last point is the paper’s central finding, and it recurs across two systems and independent pipelines: *certification has a horizon, and the horizon is set by the fitted map, not by the method*. On the Northern cod stock the certified layer has a finite horizon — six years under the two harsher floors, seven under the informative one — because the map is expansive at the reference point for every admissible carrying capacity. On the Edwards Aquifer, every positive-pumping certified kernel is empty beyond three years, an optimistic bound on a defect the out-of-sample audit exceeds. Two systems, different hydrologies, different institutions, the same shape of limitation. A certified verdict is therefore not a stronger version of a simulated one; it is a different kind of object, whose validity is *local in time* and whose locality can be quantified.

Why two systems and a cod companion. The main paper compares two methods of certification on two real assessment systems. The Northern cod analysis characterises a fitted-model harvest–protection budget; the Edwards Aquifer analysis scores governance rules under an independently fitted map. The separately titled *Part II companion: Regime Viability on the Northern Cod Stock* uses the same cod record but a different, exact-rational multiplier-bracket method. Its historical-event certificate is not a fitted-model result of this main paper.

What is found. System I: on the increasing branch of the surplus curve, the largest catch holding the reference point from itself is $C^* = g(\text{LRP}) - |e| = 91.59$ kt, and any rule’s protection margin is exactly C^* minus its catch there — at that point harvest and protection are one budget. A second constant, $g_{\max} - |e| = 215.2$ kt, separates viable-but-raised kernels from empty ones, and the two reproduce the kernel table in closed form. That identity makes the no-dominance verdict *structural, not empirical*: a reactive rule cannot out-supply a flat cap it matches in protection. System II: at the 618-ft threshold under the drought-of-record floor, training-mean pumping fails within roughly thirteen years while an interpolated 7.2% mean cut secures the threshold; the reactive rules are retained only *nominally*, on a hybrid criterion whose wet-year margin is exactly what the robust class excludes; and nothing is retained at 660 ft, where the rules are invisible to the kernel of their own trigger.

Structure. System I gives the cod harvest–protection budget; System II scores the Edwards policy family. Each keeps its source-specific methods and results. A cross-system conclusion states what certification establishes and the horizon on which it holds. The companion’s exact-rational cod analysis is separately compiled and cross-cited, rather than inserted as a second document inside this source.

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System I — The harvest–protection budget at the limit reference point

Abstract

Reference points are usually defended by simulation. Here the defence is a characterisation: on the increasing branch of the surplus curve, the robustness of a single-threshold reference point to persistent productivity shocks reduces to three constants, and the kernel computation collapses to algebra. Applied to Northern cod (NAFO 2J3KL; Schaefer fit 1983–2007; LRP 884.6 kt) under a protocol frozen before any score was computed: (1) the largest catch holding the reference point from itself is $C^* = g(\text{LRP}) - |e| = 91.59$ kt, and any rule’s protection margin is exactly C^* minus its catch there — at that point harvest and protection are one budget; a second constant, $g_{\max} - |e| = 215.2$ kt, separates viable-but-raised kernels from empty ones, and the two reproduce the kernel table in closed form. (2) That identity makes the no-dominance verdict structural, not empirical: a reactive rule cannot out-supply a flat cap it matches in protection. (3) The map is expansive at the reference point for every admissible $K \geq 2K^*$, so the certified layer has a finite horizon — six years under the two harsher floors, seven under the informative one — which is the crossing at which a geometrically growing erosion margin overtakes the worst-case trajectory, and which no catch reduction extends: it is the same for every admissible catch, including none at all. (4) Carrying capacity is not identified from above, but the functionals the results depend on are: across the profile set C^* lies in $[67.9, 95.2]$ kt — pinned to within 27 kt while K itself is not identified from above — whereas resampling the 24 transitions and refitting both parameters freely disperses the estimate over a far wider range (median 73.7 kt, 90% interval $[-89.4, 125.7]$, or 88.1 kt $([-5.6, 130.6])$ restricted to the expansive regime on which the results are conditional, 73% of replicates). The two answer different questions and the tight one is an identification result, not a precision one; 20-year survival from the reference point falls from 0.91 to 0.65 between moratorium-level and 120 kt removals. (5) Self-viability is regime-dependent: 171 kt on the post-moratorium window, 0 ± 8 kt on the modern series. On this map the reference point is protected by good years, and reactive design cannot buy protection with supply.

Keywords: northern cod; NAFO 2J3KL; surplus production; viability; robust viability kernel; harvest control rules; limit reference point

1. Introduction

After the 1992 moratorium, Northern cod (NAFO divisions 2J3KL) posed a clean governance question. Can any catch policy hold a collapsed stock’s spawning biomass above its limit reference point when productivity is depressed? Or is the reference point instead protected by good years rather than by demand management? Viability analysis supplies the natural instrument for such questions. It asks which states admit controls that keep every constraint satisfied over time, rather than which policy maximizes an objective (Aubin, 1991). It has been applied to fisheries in exactly this register: bioeconomic viability constraints (Béné, Doyen, and Gabay, 2001); viable recovery paths after collapse (Martinet, Thébaud, and Doyen, 2007); stochastic co-viability for ecosystem-based management (Doyen et al., 2012); and the numerical computation of viability kernels for fishery case studies (Krawczyk, Fox, W.W., 1970. An exponential surplus-yield model for optimizing exploited fish populations. *Transactions of the American Fisheries Society*, 99(1), 80–88. doi:10.1577/1548-8659(1970)99<80:aesmfo>2.0.co;2.

Pharo, Serea, and Sinclair, 2013; Krawczyk and Pharo, 2013). The reference point is a floor on the productive stock itself, namely the spawning biomass whose regeneration underwrites future yield. The question is therefore whether adjustments to the draw on that stock can hold the base that regenerates the yield, or whether the reference point is protected only by favourable productivity. This study answers the question by scoring rather than assertion.

Four questions organize the results. (Q1) Under which productivity regimes does any catch policy — zero catch included — hold the LRP, and where is the question vacuous because the floor exceeds the map’s maximum surplus? (Q2) Under the informative regime, what is the largest constant catch whose worst-case equilibrium remains at or above the LRP, and how do reactive rules fare against it under the declared dominance rule? (Q3) How fast does model error of the declared magnitude erode the certified safety margin when the governed map is expansive at the reference point? (Q4) Which answers are geometric properties of the rule and the threshold, which are identified properties of the fitted map, and which are sensitive to the model form or the carrying capacity?

The object is the governed surplus-production model fitted in a companion forecast-evaluation study (under separate review), which ended in a negative certificate with last-value persistence unbeaten on out-of-sample RMSE. Here the same fitted model, on the primary assessment specification (the 1983–2015 NCAM M-shift SSB series of DFO, 2016, Table A2, with LRP 884.6 kt), is scored as a closed-loop governance object. The deliverables are robust viability kernels of the LRP for a declared catch-policy family under persistent productivity-shock floors, plus a certified layer that converts the declared model defect into a safety-margin erosion. The intervention protocol — object, defect declaration, disturbance classes, policy family, and the selection rule restated in Definition 2.6 as a dominance partial order — was frozen (dated 2026-08-26) before any kernel, boundary, replay, or selection score was computed. The frozen protocol document is archived with the analysis code. Section 2 states the methods. Section 3 reports the kernels, the no-dominance verdict and the vacuous-class identity, the constructive boundary, the certified layer, and the stress replay, followed by the form comparison (Section 3.6), the carrying-capacity, stochastic, finite-duration, and uncertainty layers (Sections 3.7-3.10), and the second-specification row (Section 3.11). Section 4 discusses limitations.

Related work

Three literatures meet here: the viability analysis of managed renewable resources, the robust-viability treatment of uncertainty, and the evaluation of harvest control rules.

Viability in fisheries. That the governing question for a managed renewable resource is not which policy optimises but which states admit controls that keep every constraint satisfied over time is the founding move of viability theory (Aubin, 1991). It has been taken up in fisheries repeatedly: bio-economic viability, with the time-of-crisis function as an overexploitation indicator and irreversible configurations related to resource extinction (Béné, Doyen and Gabay, 2001); viable recovery paths toward sustainable fisheries (Martinet, Thébaud and Doyen, 2007); stochastic viability for ecosystem-based management under multiple conflicting objectives (Doyen et al., 2012); and viability as a “satisficing” alternative to optimal control, whose admissible policies are generically multiple and depend on fewer parameters than optimal or equilibrium strategies (Krawczyk and Pharo, 2013; Krawczyk et al., 2013). This paper sits in that lineage and uses its vocabulary throughout.

The production model. The governed object is a surplus-production model of Schaefer form, and the originals behind that form and behind the alternatives compared below should be named rather than assumed. The logistic surplus-production model, in which the population’s net increase is a concave quadratic in biomass with a maximum at intermediate stock size, is Schaefer’s (Schaefer, 1954); the exponential surplus-yield form fitted as a co-equal alternative is Fox’s (Fox, 1970). Both are standard, neither is contributed here, and the comparison between them is decided on residual MSE rather than on convention.

What is not standard, and is where this paper leaves the production-model literature, is the use to which the fitted form is put. That literature estimates the parameters and then asks what harvest maximises yield or profit. Here the fitted surplus enters a viability problem, and the quantity extracted from it is not an optimal effort but a *kernel*: on the increasing branch, two

constants reproduce the whole kernel table in closed form. The Schaefer-versus-Fox comparison therefore matters here for a reason it does not usually matter — the two forms give different kernels, not merely different optimal efforts — which is why it is reported as a form comparison rather than a sensitivity check.

Robust viability kernels. The set whose computation is collapsed here is the robust viability kernel: the set of initial biomasses from which at least one harvesting strategy guarantees the required production and preservation levels for all times, *whatever* the uncertainties, these being disturbances taking values in a known given set (Regnier and De Lara, 2015). Their central finding — that accounting for uncertainty sensibly shrinks the deterministic viability kernel — is the phenomenon this paper is organised around, and nothing here contradicts it.

What differs is how the set is obtained. In that literature the kernel is computed numerically, by a descending set iteration on a discretised state space. On the increasing branch of the surplus curve the computation here **collapses to algebra** instead: two constants, $C^* = g(\text{LRP}) - |e| = 91.59$ kt and $g_{\max} - |e| = 215.2$ kt, reproduce the kernel table in closed form, and every rule’s protection margin is exactly C^* minus its catch at the reference point. At that point harvest and protection are one budget. A closed-form collapse is not a faster numerical method; it is a different kind of result, and it is what makes the no-dominance verdict below structural rather than empirical.

Harvest control rules and management strategy evaluation. The standard instrument for choosing among harvest control rules is management strategy evaluation: simulate the entire management system — operating model, observation, assessment and rule — under specified uncertainties, and compare candidate rules on pre-specified performance metrics (Butterworth, 2007; Punt et al., 2016). It is a large, mature and practically indispensable machinery. The opening remark of this paper, that reference points are usually defended by simulation, is a description of that literature rather than a criticism of it.

The distinction is in what the two deliver, and it can be stated without rhetoric. An MSE verdict is *conditional on the operating model and on the ensemble simulated*: a no-dominance finding holds for the rules, the uncertainties and the replicates that were actually run. The no-dominance verdict here falls out of an identity — protection margin equals C^* minus catch — and therefore holds for *every* rule in the class at once, including rules that were never simulated. That is a genuine difference in kind.

It is also a narrower one, and the narrowing should be conceded. MSE routinely handles multiple conflicting objectives, implementation error, observation error and institutional constraints simultaneously; the scalar reduction here buys its closed form by giving most of that up. The two are complements rather than rivals: the characterisation says what cannot be bought at any price within the class, while the simulation says which rules do best on the metrics that were specified.

2. Methods

2.1 Operating model and data

All inputs are public and the analysis is reproducible: every stochastic layer uses a fixed seed and a fixed draw budget (Sections 3.8 and 3.10), and every deterministic layer is a closed-form or fixed-point computation. The stock series is the Northern cod (NAFO 2J3KL) spawning-stock biomass of DFO (2016), Table A2. Annual landings are Schijns et al. (2021), Table 1. The fit window is 1983–2007 (24 one-step transitions; the out-of-sample audit runs 2008–2015), and the protocol was frozen on 2026-08-26, before any kernel, boundary, replay, or selection score was computed. The training-window residuals of the fitted map have mean -10.9 kt, SD 114.9 kt, range $[-329.0, +206.6]$ kt, and lag-1 autocorrelation 0.55 . The 1992 transition residual is the minimum (-329.0 kt). The training-window maximum stock is 940.75 kt, which is why the fitted carrying capacity, pinned at its 5000 kt optimization bound, sits above the observed

range (the declared optimization box runs from just above the training maximum to 5000 kt). A companion forecast-evaluation study scored the same point-estimate map against persistence on the same series and issued a negative certificate (persistence unbeaten out of sample). That certificate is scored there, not here, and the closed-loop object below is the same fitted map by construction.

All analyses beyond the frozen kernel tables — the carrying-capacity grid, the stochastic viability layers, the finite-duration floors, and the bootstrap bands of Sections 3.7–3.10 — are additional scored objects executed after the freeze on the declared machinery. None of them replaces or alters the frozen family, the frozen floor classes, or the dominance rule of Definition 2.6.

2.2 The governed object and the scoring rule

The governed object is the companion forecast-evaluation study’s own stock-flow class.

$$S_{t+1} = [S_t + rS_t(1 - S_t/K) - C_t + e_t]_+$$

The Allee term is off. The map is fitted by one-step least squares on 1983–2007 with annual landings (Schijns et al., 2021), giving $r = 0.2369$ and $K = 5000$ kt (pinned at its optimization bound; the series never approaches carrying capacity, a declared defect; the LRP-boundary results depend chiefly on the identified r). Fit residual SD is 114.9 kt. The defect declaration is $\varepsilon = 329.0$ kt yr⁻¹ (the 1992 collapse transition). The out-of-sample audit on 2008–2015 gives a maximum of 47.1 kt, which does not exceed the declared defect (by contrast, the groundwater object of the companion intervention study exceeds its declared defect out of sample).

The declared formal objects are stated as definitions and propositions so that the empirical content can be read off cleanly.

Definition 2.1 (Governed surplus-production map). The governed object is the one-step map above with parameters $(r, K) = (0.2369, 5000 \text{ kt})$ fitted by one-step least squares on 1983–2007, Allee term off, and the declared defect declaration $\varepsilon = 329.0$ kt yr⁻¹.

Definition 2.2 (Safe set). The single declared threshold is $K^* = \text{LRP} = 884.6$ kt (the 1983–1989 mean of Table A2). The safe set is $[K^*, \infty)$. No row on the second assessment specification (the 1954–2024 extended xteNCAM series, LRP 276 kt) enters the frozen primary protocol. The labelled post-freeze row of Section 3.11 reports that specification separately, unpooled, against its own reference point, and no verdict transfers between the two.

Definition 2.3 (Disturbance classes). The disturbance classes are persistent additive productivity floors from the fit-window residual distribution: (i) the perpetual worst observed one-step shock (-329.0 kt yr⁻¹), (ii) the 5th-percentile class (-287.4 kt yr⁻¹), and (iii) the 10th-percentile class (-80.9 kt yr⁻¹). Because this object has no independent input channel (unlike the recharge series of the companion groundwater study), the disturbance classes and the defect declaration are the same measured quantity in two roles.

Definition 2.4 (Governance family). The governance family consists of the following declared policies. BAU has $C \equiv 5$ kt (moratorium-level inshore removals, the declared implementable use post-1992). Flat caps are $\rho \cdot 240$ kt with $\rho \in \{1.0, 0.75, 0.5, 0.25, 0.0\}$ (i.e. 240, 180, 120, 60, 0 kt; every member is scored). S1 is the DFO-2009 critical-zone rule (DFO, 2009) at a declared 60 kt cap (60 above the LRP, 0 below). The cascade is 60/30/5/0 kt at LRP/0.75·LRP/0.5·LRP/below; the sub-LRP stages are declared scenarios, not verified institutions. Two new families are reactive in the strict sense — their catch is a function of the current stock, switching above the LRP and vanishing at or below it. Family A is surplus-proportional: $C(S) = \phi g(S)$ for $S \geq \text{LRP}$ and 0 below, with $\phi \in \{0.25, 0.50, 0.75\}$; its catch scales with the harvested surplus $g(S) = rS(1 - S/K)$ and therefore feeds back on the stock. Family B is graded above the LRP: graded2 is 0 / LRP–1.25·LRP: 60 / >1.25·LRP: 90 kt, and graded3 is 0 / LRP–1.15·LRP: 30 / 1.15·LRP–1.35·LRP: 60 / >1.35·LRP: 90 kt. Families A and

B are post-freeze scored objects added to address the reactive-management question directly; they are not declared institutions.

Two further reactive designs were scored and are not reported as a family because they are dominated. The surplus-harvest fraction $C(S) = \alpha \cdot \max(0, g(S) - g(K^*))$ removes only the surplus above the LRP’s own regeneration, and the reactive ramp $C(S) = \min(C_{\max}, q(S - K^*))$ ramps linearly above the LRP. Under the declared source-year classes, every member of both was empty at the 5th-percentile $T = \infty$ reading (the clause-H1 failure of Result 3.1 applies to them as to every positive-catch rule) and, on the training-supply replay, harvested at or below the moratorium level (0.18–0.71 kt mean catch for the surplus-harvest fractions, 0.64 kt for the ramp; 5 kt for BAU), so neither family achieves the “harvest more than the moratorium” property that lets Family A register a genuine trade-off. They are recorded as rejected designs and omitted from the kernel tables below.

Definition 2.5 (Robust viability kernel, closed-loop reading). For each policy and disturbance class, the robust viability kernel is the set of initial SSB values from which the closed-loop path keeps the stock at or above the LRP under the persistent floor. Table 1 reports kernel lower boundaries at $T = 1$ and $T = \infty$, with “empty” denoting that no state is robustly viable. The term is used in its closed-loop reading — the robust positively invariant set of a fixed declared policy — not in the classical viability reading of Aubin (1991), which is existential over controls. No control choice enters the kernel computation; only the evaluation of declared rules under the disturbance floor enters. The word is retained because the object answers the same question (“from which states can the constraint be held”) in the policy-fixed form.

Definition 2.6 (Dominance partial order). The rule the frozen protocol calls its retention rule is restated here, clauses verbatim, as a partial order on policies rather than a filter that selects an adopted policy: a non-BAU policy *dominates* BAU only if all three of the following hold.

- (H1) Its kernel is at least as protective as BAU’s at every reading (compared on the kernel lower boundary; empty = worst).
- (H2) It improves on BAU somewhere.
- (H3) At some reading where it improves, its mean allowed catch exceeds that of every at-least-as-protective flat cap.

Why the filter vocabulary is not used. Clause (H1), read at every reading with empty scored as worst, is a structural block rather than a comparison that could go either way: under the 5th-percentile class at $T = \infty$ every declared positive-catch rule has an empty kernel while BAU’s is nonempty (2219.6 kt), so within the declared family no positive-catch rule can satisfy (H1) at all — the rule’s outcome is fixed by the declared classes and BAU’s marginal nonemptiness there. Presenting that outcome as a “retention” decision would overstate it; the vocabulary here is dominance, the outcome is stated as a selection property of the declared partial order (Result 3.1), and the two identity-level statements of the construction (the vacuous-class identity and the structural block itself) are stated as definitional notes.

Definition 2.7 (Certified layer conversion). The certified layer applies the defect-to-margin conversion in the form the fitted map admits. If the closed loop contracts with rate $a < 1$ on the safe domain, the erosion margin is $r_T = \varepsilon(1 - a^T)/(1 - a)$. Otherwise the expansive form $r_T = \varepsilon(a_{\max}^T - 1)/(a_{\max} - 1)$ applies with $a_{\max} = \sup |F'|$ over the safe domain, and the certified kernel is the nominal kernel of $K^* + r_T$.

2.3 The catch-timing convention

One modelling choice has to be declared explicitly, because it moves the reported numbers. Write the one-step residual of the governed map as

$$e_t = S_{t+1} - (S_t + g(S_t) - C_{t+\delta}), \quad \delta \in \{0, 1\},$$

so that $\delta = 0$ pairs the transition $t \rightarrow t + 1$ with the catch of the year the transition leaves (*source-year*) and $\delta = 1$ pairs it with the catch of the year it enters (*destination-year*). The two are different residual distributions on one and the same fitted map: at the committed (r, K) the source-year pool has mean -10.9 kt, SD 114.9 kt and lag-1 autocorrelation 0.554 , against -20.4 kt, 135.0 kt and 0.652 for the destination-year pool, and the classes of Definition 2.3 move with them ($-329.0/-287.4/-80.9$ kt against $-460.0/-318.8/-114.9$ kt).

Convention (adopted). Every kernel, boundary, floor, replay, figure and table of this paper is computed in the source-year convention ($\delta = 0$).

It is adopted on three grounds, none of them provenance. (i) *Coherence*. The committed parameters are the minimisers of the source-year loss. The estimation routine regresses the increment $S_{t+1} - S_t$ on the catch of year t , so $r = 0.2369$ and $K = 5000$ kt minimise the mean squared source-year residual: $12,772$ kt², against $17,873$ kt² for the destination-year pool at those same parameters (28.5% lower). Scoring those parameters against destination-year residuals would report classes drawn from a loss the fit never minimised. (ii) *Fit*. The source-year pool is the better-fitting and the less serially correlated of the two, as tabulated above. (iii) *No interior alternative*. Refitting (r, K) freely against the blended catch $wC_t + (1 - w)C_{t+1}$ gives a mean squared error monotone decreasing in w over the whole unit interval, with the optimum at the boundary $w = 1$: $17,713$ kt² at $w = 0$ (where the refit gives $r = 0.2084$), $15,026$ kt² at $w = 0.5$, and $12,772$ kt² at $w = 1$; the out-of-sample error over 2008–2015 moves the same way (834.9 to 783.1 kt², eight one-step transitions). There is no interior optimum, so a half-and-half compromise is not supported by the data.

Limitation, stated because a referee will ask. Nothing in the archived source documentation records whether the spawning-stock column of DFO (2016) Table A2 is measured before or after the calendar year’s removals, and the biology does not settle it either: Northern cod spawn in March–May, so an SSB observation sits mid-year while landings are calendar-year totals, and the interval between two SSB observations spans the back half of one year’s catch and the front half of the next. On that reading neither pure convention is mechanistically exact and the truth is a blend — which is exactly why ground (iii) is needed. The convention is therefore declared on statistical and coherence grounds rather than derived from provenance, and it is stated here rather than left implicit. The sensitivity is not suppressed: the primary kernels and their boundary tables were also computed under $\delta = 1$, and those artifacts are archived with the superseded runner rather than overwritten, so every affected number can be recomputed from the archive instead of taken on trust. Four quantities move materially — the constructive bound (91.59 kt against 57.61 kt), the number of vacuous floor classes (one against two), the certified horizon ($6/7$ years against $5/6$), and the 60-kt rules’ $T = \infty$ boundary (884.6 kt against 900.3 kt). No parameter estimate, no form comparison and no certificate direction changes.

2.4 What the kernel problem reduces to

The kernels of Section 3 are computed by exact backward recursion on the declared interval engine, but they are not merely numerical. On the increasing branch of the surplus curve — the only branch the safe set occupies here, since $K^* < K/2$ — the whole table reduces to three constants. Write g for the surplus, $|e|$ for the persistent floor, $C(S)$ for the policy’s catch and $F(S) = S + g(S) - C(S) - |e|$ for the worst-case closed loop. Every statement below is verified cell by cell against the committed kernels by `campaign_e2_structure_v3.py` (23 checks).

Proposition 2.1 (Constructive bound and the protection–supply budget). For any policy whose catch at the reference point is $C(K^*)$, the safe set is held from itself if and only if $C(K^*) \leq C^* := g(K^*) - |e|$, and the margin by which it is held is exactly $C^* - C(K^*)$.

Proof. One step from K^* gives $K^* + g(K^*) - C(K^*) - |e|$, which is at least K^* under exactly the stated inequality, and the margin is the difference of the two sides. It remains only that the reference point is the binding state, i.e. that $S \mapsto S + g(S) - C(S)$ attains its minimum over the safe set at K^* : this holds on the increasing branch for every declared policy, the graded rules included, because their catch is piecewise constant and each upward jump (30 kt at most) is smaller than the surplus gained across the jump. The campaign verifies it directly for all twelve policies. $\square$

The largest catch that holds the reference point from itself is therefore $C^* = 172.46 - 80.87 = 91.59$ kt (Section 3.3). The identity is the substantive part: at the reference point harvest and protection are one budget, and every kilotonne taken there is a kilotonne of margin surrendered. Section 3.2 shows that this, and not the tabulation, is why no declared rule dominates the moratorium.

Proposition 2.2 (Perpetual bound and the infinite-horizon boundary). For a constant catch C , the $T = \infty$ kernel is $[b_\infty, \infty)$ with $b_\infty = \max(K^*, s_-(C))$, where $s_-(C)$ is the smaller root of $g(S) = C + |e|$; it is empty if and only if $C > C_{\text{vac}} := g_{\text{max}} - |e|$.

Proof. $F(S) - S = g(S) - C - |e|$ is positive exactly between the two roots of $g(S) = C + |e|$. From any state above the smaller root the trajectory increases towards the larger and so never falls below the smaller; from any state below it the trajectory declines. If $C + |e| > g_{\text{max}}$ there is no root and the trajectory declines everywhere. $\square$

Two constants therefore organise the last column of Table 1: below $C^* = 91.59$ kt the kernel is the whole safe set; between C^* and $C_{\text{vac}} = 296.09 - 80.87 = 215.2$ kt it is a proper half-line strictly above the reference point; above C_{vac} it is empty. The column is reproduced in closed form to 0.1 kt for all six flat caps.

Proposition 2.3 (Reactive families). For $C(S) = \phi g(S)$ above the LRP and 0 below, the closed loop is $F(S) = S + (1 - \phi)g(S) - |e|$ and there are three regimes: the kernel is the whole safe set iff $\phi \leq 1 - |e|/g(K^*)$; it is a proper half-line iff that fails but $\phi < 1 - |e|/g_{\text{max}}$; and it is empty iff $\phi \geq 1 - |e|/g_{\text{max}}$.

Proof. Proposition 2.1 with $C(K^*) = \phi g(K^*)$, and Proposition 2.2 with the effective surplus $(1 - \phi)g$. $\square$

Under the declared 10th-percentile floor the two thresholds are 0.531 and 0.727. The members $\phi = 0.25$ and 0.50 hold the entire safe set — the second by 5.4 kt, not by the wide margin an earlier version of this paper reported (Section 3.4) — $\phi = 0.75$ is empty, and the band $0.531 < \phi < 0.727$ is the regime in which the rule is viable but only from above the reference point. No member of the originally declared family fell in that band, which is why the erroneous criterion went unnoticed; the family has been extended by $\phi = 0.60$ so that Table 1 exhibits it.

Proposition 2.4 (The certified horizon is a crossing). For a constant catch — the case the certified layer is computed on — let S_{hi} be the domain ceiling and $r_T = \varepsilon(a_{\text{max}}^T - 1)/(a_{\text{max}} - 1)$ the erosion margin of Definition 2.7. The certified kernel at horizon T is nonempty iff $F^T(S_{\text{hi}}) \geq K^* + r_T$, and since $F^T(S_{\text{hi}})$ is nonincreasing in T while r_T increases, the certified horizon is the unique crossing $T^* = \max\{T : F^T(S_{\text{hi}}) \geq K^* + r_T\}$. It is finite whenever the loop is expansive.

Proof. For a constant catch $F'(S) = 1 + g'(S) > 0$ throughout $[K^*, S_{\text{hi}}]$ — the surplus slope reaches -1 only at $S = \frac{K}{2}(1 + 1/r) \approx 13,000$ kt, beyond the domain — so F is increasing and the supremum over initial states of the T -step minimum is attained at the ceiling; the rest is the definition of the certified kernel. Finiteness follows because r_T grows geometrically in T while $F^T(S_{\text{hi}})$ is bounded below (it converges to the attracting fixed point when one exists, and declines otherwise). $\square$

Nothing in this subsection uses the Schaefer form. The only properties invoked are that the surplus is unimodal with its peak above the reference point and that the safe set lies on the increasing branch, so the constants travel to any surplus function with those two properties. They do: on the Fox row of Section 3.6 the formula $C^* = g(K^*) - |e|$ gives 79.1 kt against the tabulated 79.05, with $C_{\text{vac}} = 111.2$ kt and $g_{\text{max}} = 192.0$ kt; on the two Allee rows it gives 123.27

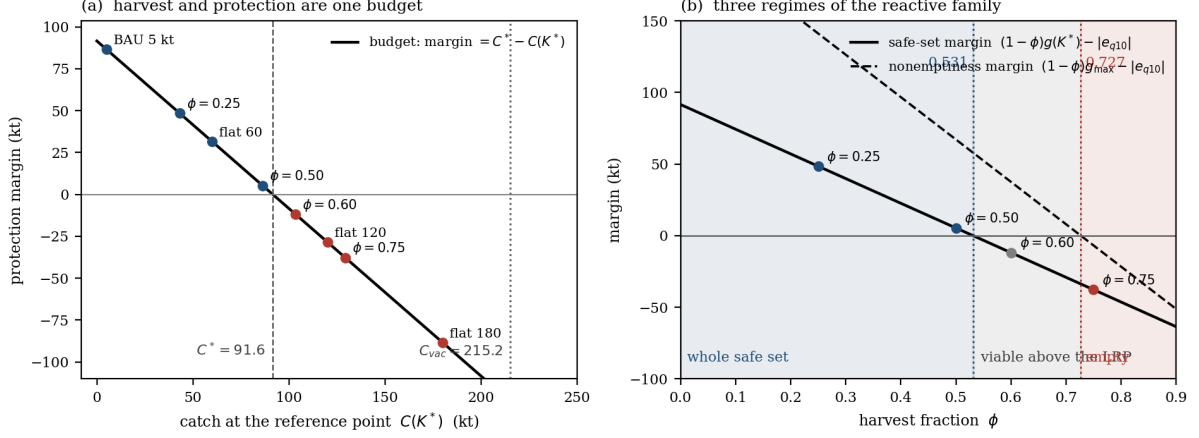


Figure 1: The two structural results of Section 2.4. (a) The protection–supply budget of Proposition 2.1: every policy lies on the line, so a rule’s harvest at the reference point and the margin by which it holds that point sum to $C^* = 91.59$ kt; the flat 60-kt cap, BAU and $\phi \leq 0.50$ sit on the protective segment, the 120- and 180-kt caps and $\phi = 0.75$ below it, and the dotted line at $C_{\text{vac}} = 215.2$ kt is where the kernel becomes empty. (b) The three regimes of Proposition 2.3: the safe-set margin $(1 - \phi)g(K^*) - |e|$, zero at $\phi = 0.531$, and the nonemptiness margin $(1 - \phi)g_{\text{max}} - |e|$, zero at $\phi = 0.727$, with the four declared members marked.

kt (declared s_0) and 115.19 kt (data-preferred), both reproducing the tabulated constructives exactly. The Schaefer numbers of Section 3.3 are one instance of a statement that holds across the forms the paper fits.

The crossing reproduces the computed horizons exactly — $T^* = 6$ under the perpetual-worst and 5th-percentile floors and $T^* = 7$ under the informative one, for both BAU and zero catch (Section 3.4). Where the loop has an attracting fixed point $S^\dagger$ the crossing is bounded below by $\lfloor \log(1 + (S^\dagger - K^*)(a_{\text{max}} - 1)/\varepsilon) / \log a_{\text{max}} \rfloor$, which is attained once the ceiling trajectory has converged: it gives 7 against 7 under the informative floor, and 4 against 6 under the 5th-percentile floor, where at six years the ceiling trajectory is still 1,500 kt above the attractor.

3. Results

3.1 Robust kernels (nominal)

Table 1. Lower boundaries of the robust viability kernels of the LRP (kt) under the declared catch-policy family (frozen rules plus the two post-freeze reactive families) and disturbance classes, in the source-year convention. “empty” denotes that no state is robustly viable at that horizon. The infinite-horizon emptiness of Section 3.2 is recorded in the table, not only in prose.

Schaefer, M.B., 1954. Some aspects of the dynamics of populations important to the management of the commercial marine fisheries. <i>Bulletin of the Inter- American Tropical Tuna Commission</i> , 1(2), 27–56. Reprinted in <i>Bulletin of Mathematical Biology</i> , 53, 253–279, 1991, doi:10.1007/bf02464432.						
Policy	Worst, $T = 1$	$T = \infty$	q05, $T = 1$	$T = \infty$	q10, $T = 1$	$T = \infty$
BAU (5 kt)	1025.5	empty	989.0	2219.6	884.6	884.6
flat 240 kt	1233.5	empty	1196.4	empty	1014.0	empty
flat 180 kt	1180.0	empty	1143.1	empty	961.5	1637.8
flat 120 kt	1126.8	empty	1090.1	empty	909.3	1082.3
flat 60 kt	1073.8	empty	1037.2	empty	884.6	884.6
S1 / cascade	1073.8	empty	1037.2	empty	884.6	884.6
flat 0 kt	1021.1	empty	984.7	2070.9	884.6	884.6
Family A, $\phi=0.25$	1064.7	empty	1027.3	empty	884.6	884.6
Family A, $\phi=0.50$	1111.3	empty	1071.7	empty	884.6	884.6
Family A, $\phi=0.60$	1131.2	empty	1091.0	empty	895.2	1074.8
Family A, $\phi=0.75$	1161.0	empty	1119.9	empty	920.2	empty
Family B, graded2	1073.8	empty	1037.2	empty	884.6	884.6
Family B, graded3	1073.8	empty	1010.9	empty	884.6	884.6

Under the 10th-percentile class the moratorium, zero catch, the flat 60-kt cap, the critical-zone rule, the cascade, both graded rules, and the surplus-proportional family at $\phi \leq 0.50$ all hold the entire safe set $[884.6, 10^4]$ kt at every horizon. Every higher nonzero cap lifts the kernel’s lower edge above the LRP, and the column is not a list of accidents: by Proposition 2.2 the boundary is $\max(K^*, s_-(C))$ with $s_-(C)$ the smaller root of $g(S) = C + |e_{q10}|$, so the catches divide at $C^* = 91.59$ kt (whole safe set), $C_{\text{vac}} = 215.2$ kt (nonempty) and beyond it

(empty) — which is exactly what the last column shows, to 0.1 kt, for all six flat caps. Under the perpetual-worst floor the infinite-horizon kernel is empty for every policy, because the maximum surplus $g_{\max} = rK/4 = 296 \text{ kt yr}^{-1}$ lies below that floor. Under the 5th-percentile class the infinite-horizon kernel is nonempty for the moratorium (2219.6 kt) and zero catch (2070.9 kt) but empty for every positive-cap rule.

The critical-floor axis makes the qualification explicit. At zero catch, $\bar{e} = g_{\max} = 296.1 \text{ kt yr}^{-1}$ separates vacuous from informative classes. Only the perpetual-worst floor (-329.0 kt) sits beyond it and is vacuous. The 5th-percentile (-287.4 kt) and 10th-percentile (-80.9 kt) floors both lie on the informative side, which is why the constructive boundary of Section 3.3 exists for the 10th-percentile class and the 5th-percentile class now carries informative, non-vacuous content.

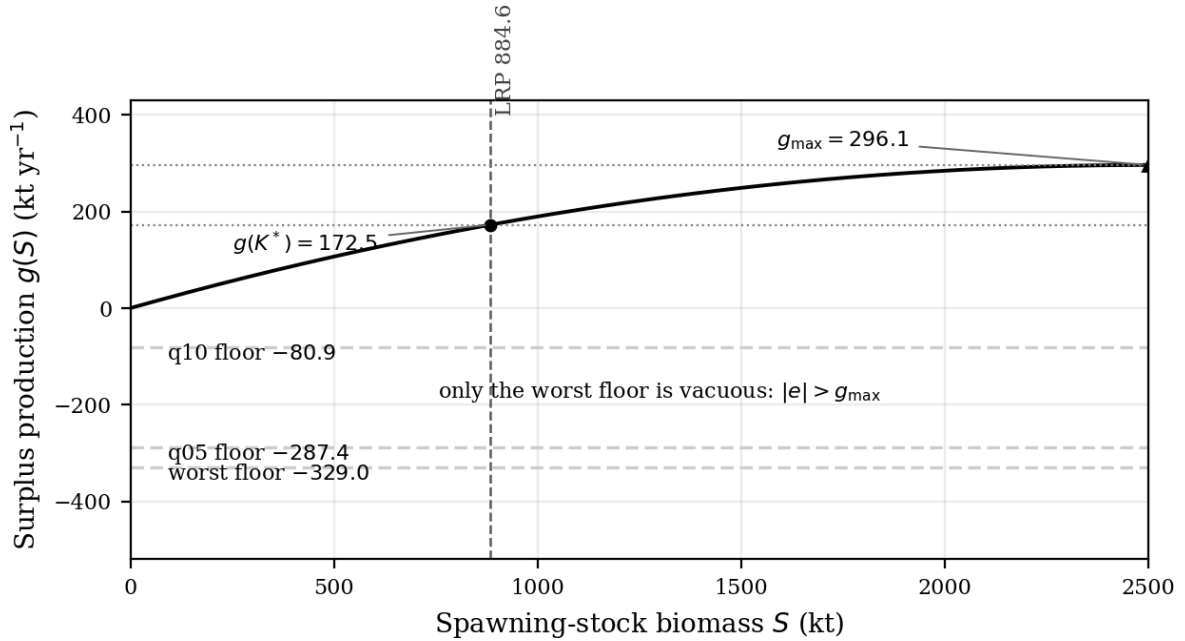


Figure 2: Surplus production of the registered fit with the three persistent floors; only the perpetual-worst floor sits beyond $g_{\max} = 296.1 \text{ kt yr}^{-1}$, so only that floor’s emptiness is vacuous.

3.2 The no-dominance verdict and the vacuous-class identity

The first result concerns dominance; the second is the vacuous-class identity, stated as a definitional note. Both are scoped to the declared rule and disturbance classes.

Result 3.1 (No dominance). Under the dominance partial order of Definition 2.6 (the frozen rule; clauses verbatim), no non-BAU policy dominates BAU.

Reason. The binding failure is clause (H1), read at its stated scope: “at every reading,” with empty scored as worst. Under the 5th-percentile class at $T = \infty$ every positive-catch rule — the flat caps, the critical-zone rule, the cascade, both graded members of Family B, and every member of the surplus-proportional Family A — has an empty kernel, while BAU’s kernel is nonempty there (2219.6 kt). Because empty is worst, none of them is at least as protective as BAU at that reading, so clause (H1) fails for every positive-catch rule. The informative 10th-percentile class is not where the verdict is decided: under that class the flat 60-kt cap, the critical-zone rule, the cascade, and both graded rules hold the LRP from itself at every horizon (884.6 kt at $T = 1$ and $T = \infty$), exactly matching BAU, and the surplus-proportional family at $\phi \leq 0.50$ holds it too while harvesting more than the moratorium — so the informative class is where the reactive family *almost* passes, not where it fails. Clause (H3) is therefore a second,

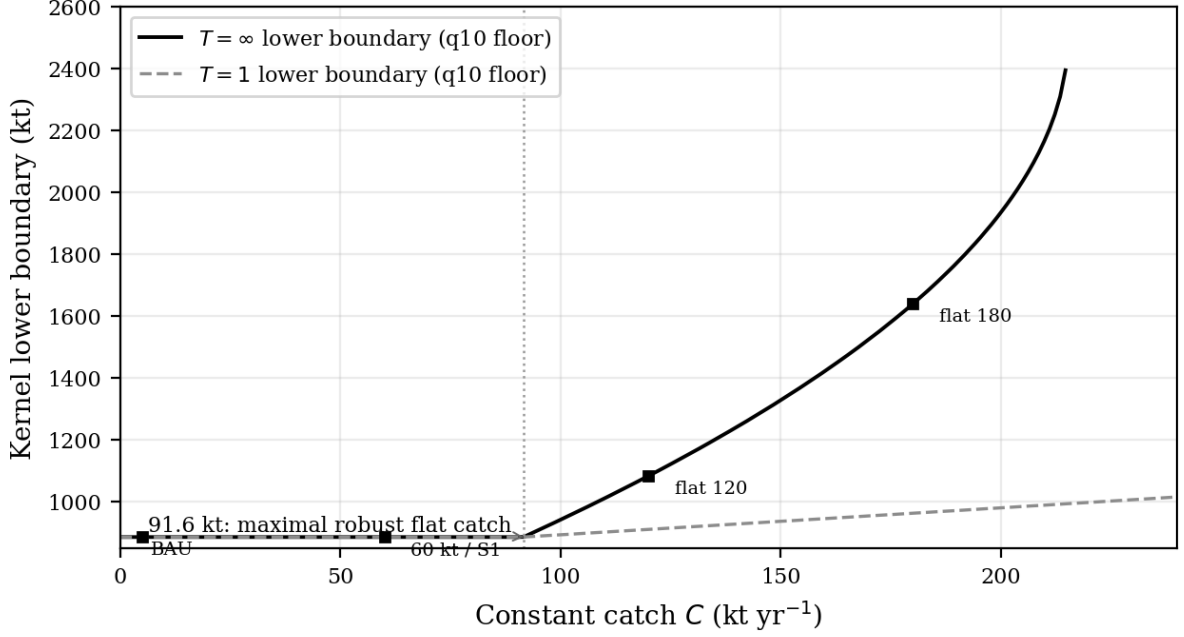


Figure 3: Kernel lower boundary versus constant catch under the 10th-percentile class: the $T = \infty$ boundary leaves the LRP at the constructive 91.59 kt.

independent failure rather than the mechanism: the equal-protective flat-60 cap (mean allowed catch 60 kt) exceeds the training means of the critical-zone rule (10.0 kt), the cascade (16.3 kt), graded2 (10.0 kt), and graded3 (5.0 kt). At $\phi = 0.75$ Family A also fails on the informative class itself, its $T = \infty$ kernel empty (920.2 kt at $T = 1$ against 884.6; see Result 3.4). No non-BAU policy therefore dominates BAU, and the mechanism is the empty-at-the-5th-percentile-class reading of clause (H1), with the supply comparison as a second failure. The verdict’s status is selection-theoretic, not empirical: given the declared classes, clause (H1) blocks every positive-catch rule by construction (Definition 2.6’s note), so the no-dominance outcome is a property of the declared partial order and classes as much as of the policies. The companion groundwater evaluation operates under a different constraint structure — not a single fatal floor — so its retained reactive rules and this no-dominance outcome are not comparable selections. $\square$

Which governance architecture justifies its additional structure is system-dependent. The framework’s deliverable is the scored comparison, not a universal architecture verdict.

Definitional note 3.2 (The vacuous-class identity). Under the perpetual-worst floor, no catch policy — zero catch included — holds the LRP. This statement is a definitional note rather than a numbered result: it is an arithmetic identity of the declared class, not an empirical finding, and it is not numbered on a par with the constructive boundary or the stochastic layer. Under the 5th-percentile and 10th-percentile classes the statement is not vacuous: those floors lie below the map’s maximum surplus and the classes carry informative content.

Statement. The perpetual-worst floor (-329.0 kt yr $^{-1}$) exceeds the map’s maximum surplus ($g_{\max} = 296$ kt yr $^{-1}$). Every trajectory therefore declines for every catch, zero included: $|e| > g_{\max}$ implies monotone decline for every $C \geq 0$ — an identity of the map’s algebra, not a measurement. The 5th-percentile class (-287.4 kt yr $^{-1}$) and the 10th-percentile class (-80.9 kt yr $^{-1}$) both lie below $g_{\max}$ and are not vacuous: their kernels carry substantive content (Table 1, Result 3.3), and the 5th-percentile moratorium kernel is nonempty (2219.6 kt). The correction of the residual convention therefore reduces the vacuous family from two classes to one. The analogy to the companion groundwater institutional certificate is one of form only; that certificate is institutional, this one is a floor-above-surplus identity, and the two are not pooled. $\square$

On this map the reference point is protected by good years rather than by demand management — but under the corrected class the margin that good years must supply is smaller than the frozen reading implied.

3.3 Constructive boundary

The maximal robust flat catch is the largest constant catch whose worst-case low equilibrium stays at or above the LRP. Under the 10th-percentile class this constructive boundary is **91.59 kt**.

Result 3.3 (Constructive boundary). Under the 10th-percentile class, the maximal robust constant catch is 91.59 kt yr⁻¹. Under the 5th-percentile and perpetual-worst classes no positive catch is robust.

Reason. The value is $g(K^*) - |e_{q10}| = 172.46 - 80.87 = 91.59$ kt yr⁻¹. It is independent of the family scaling and rests only on the identified r and the corrected 10th-percentile floor. Under the 5th-percentile floor the corresponding quantity is $172.46 - 287.36 = -114.9$ kt, so no constant catch is robust there; under the perpetual-worst floor the same holds ($172.46 - 328.97 = -156.5$ kt). $\square$

Proposition 2.1 gives the bound a second reading: the margin by which any policy holds the reference point is exactly 91.59 kt minus its catch there, so the bound is the whole budget and harvest and protection are the same quantity seen from two sides. This is certification geometry at one declared shock class, not a harvest rule. The stochastic analogue of the bound is reported in Section 3.8, and its sampling distribution in Section 3.10. The corrected bound is read as order 70–90 kt, not the 40–60 kt of the frozen convention.

Result 3.4 (Reactive rules above the LRP). The surplus-proportional family is genuinely reactive: at $\phi = 0.25$ and 0.50 it holds the entire safe set under the 10th-percentile class at every horizon, including $T = \infty$, and its catch feeds back on the stock (it rises with the surplus and vanishes below the LRP). It is nonetheless empty at the 5th-percentile $T = \infty$ reading, so its protection is a property of the informative class only — a trade-off, not a strict improvement over BAU, as set out in Result 3.1. The graded family is protective on the informative class but not supply-superior. The two families are compared against the frozen rules in Table 1.

Reason. For a surplus-proportional rule $C = \phi g(S)$ the closed loop is $F(S) = S + (1 - \phi)g(S) + e$, a single concave quadratic, and Proposition 2.3 gives three regimes rather than two. The safe set is held from itself iff the scaled surplus at the reference point covers the floor, $(1 - \phi)g(K^*) \geq |e_{q10}|$, i.e. $\phi \leq 1 - 80.87/172.46 = 0.531$; at $\phi = 0.25$ (0.50) the margin is 48.5 (5.4) kt, so both qualify, the second only just. Nonemptiness is the weaker condition $(1 - \phi)g_{\max} > |e_{q10}|$, i.e. $\phi < 1 - 80.87/296.09 = 0.727$, and it fails at $\phi = 0.75$ (margin -37.8 kt), so that member's $T = \infty$ kernel is empty (920.2 kt at $T = 1$). Between the two thresholds the rule is viable but only from above the reference point: at the added member $\phi = 0.60$ the informative kernel is $[1074.8, 10^4]$ kt, not the safe set, and it is this regime that an earlier version of this paper missed by quoting 0.727 as the threshold for holding the safe set. The three tabulated verdicts were unaffected, which is why the error survived every numeric check. Under the 5th-percentile floor the criterion is $\phi < 0.029$, so no surplus-proportional rule with positive ϕ holds that class — which is exactly why the same family is empty there, and why it does not dominate BAU under the frozen rule. Under the perpetual-worst floor the loop declines for every ϕ , so the kernel is empty. The graded members (Family B) are piecewise-constant and are scored exactly as the frozen rules: under the 10th-percentile class both hold the LRP from itself at every horizon, and both are empty at the 5th-percentile $T = \infty$ reading. The rejected-designs note following Definition 2.4 reports the same clause-H1 emptiness for the surplus-harvest and ramp designs, which moreover do not improve supply. Supply replays (training mean allowed catch over the observed states) give BAU 5 kt, S1 10.0 kt, the cascade 16.3 kt, flat-60 60 kt, flat-0 0 kt, graded2 10.0 kt, graded3 5.0 kt, and the surplus-proportional family 7.4/14.7/22.1 kt at $\phi = 0.25/0.50/0.75$. The critical-zone rule's low mean is a property of the window, not of

the rule: the cut is active in 20 of the 24 replay years (the stock sat below the reference point in all but the 1985 and 1987–1989 states, 83% of the window), a fact about the collapsed-era estimation window rather than about the rule’s post-recovery supply properties, and the two regimes are not mixed. $\square$

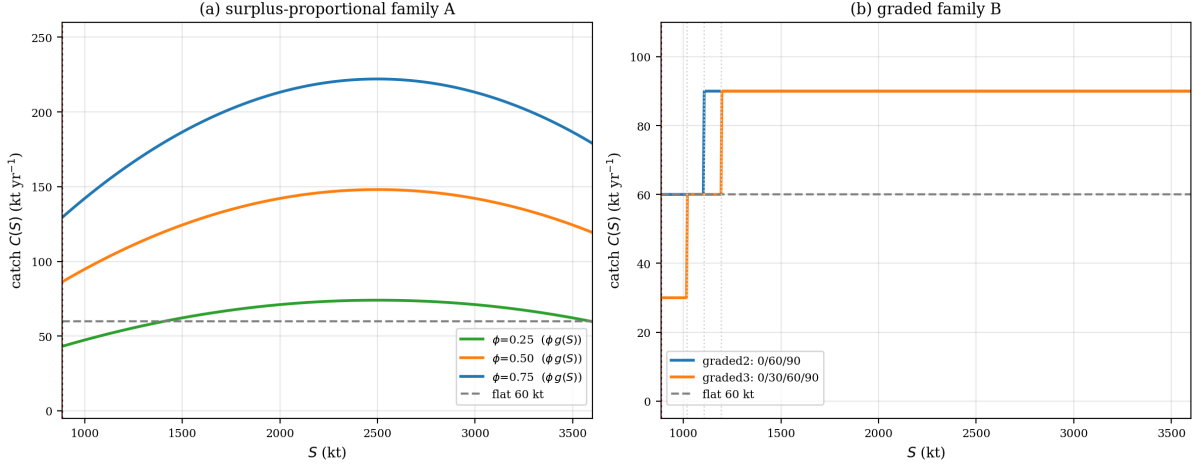


Figure 4: The two reactive families: (left) the catch schedules of the surplus-proportional Family A ($C = \phi g(S)$, $\phi = 0.25, 0.50, 0.75$) against the flat 60-kt cap, showing that the rule harvests less than the cap near the LRP while scaling with the surplus; (right) the catch schedules of the graded Family B (graded2 and graded3) against the flat 60-kt cap. Both families vanish at or below the LRP. The dominated surplus-harvest ($\alpha \cdot \max(0, g(S) - g(K^*))$) and reactive-ramp designs are recorded in the rejected-designs note following Definition 2.4 and not plotted.

3.4 Certified layer: the expansion obstruction

The conversion of Definition 2.7 needs the closed loop’s contraction rate. The next result shows that no such contraction is available at the declared safe set, and that the certified layer therefore has a finite horizon: six years under the two harsher floors, seven under the informative one.

Result 3.5 (Expansion obstruction). On the governed surplus-production map of Definition 2.1 the certified kernel is empty from $T = 7$ under the perpetual-worst and 5th-percentile floors, and from $T = 8$ under the 10th-percentile floor, for every declared policy, zero catch included. At the last nonempty horizon the certified sets lie far above the observed range of the stock: $[5132.9, 10^4]$ kt (perpetual worst, BAU, $T = 6$), $[4593.2, 10^4]$ kt (5th percentile, BAU, $T = 6$), and $[4560.3, 10^4]$ kt (10th percentile, BAU and zero catch, $T = 7$); under the informative floor the 180- and 240-kt caps are already empty at $T = 7$.

Reason. Here $F'(S) = 1 + r(1 - 2S/K)$ is increasing as S falls. At the LRP, $F'(K^*) = 1.153 > 1$. The governed surplus map is therefore expansive at the declared safe set, contracting only above $K/2 = 2500$ kt — a stock level the series never approaches. The contraction form of the conversion is therefore inapplicable. The expansive form $r_T = \varepsilon(a_{\max}^T - 1)/(a_{\max} - 1)$ grows without bound; with the committed $\varepsilon = 328.9725$ kt — the perpetual-worst class of Section 2.3, printed there to one decimal as 329.0 — and $a_{\max} = 1.1531$: $r_1 = 329.0$, $r_2 = 708.3$, $r_3 = 1145.7$, $r_4 = 1650.1$, $r_5 = 2231.7$, $r_6 = 2902.4$, $r_7 = 3675.7$, $r_8 = 4567.4$ kt, so the shifted threshold $K^* + r_T$ runs 1213.6, 1592.9, 2030.3, 2534.7, 3116.3, 3787.0, 4560.3, 5452.0 kt at $T = 1, \dots, 8$.

Emptiness is not simply the shifted threshold outgrowing the carrying capacity: at $T = 7$ the threshold (4560.3 kt) is still below $K = 5000$ kt, and the certified kernel is empty under the two harsher floors nevertheless. What fails is the trajectory, not the arithmetic of the shift. Under the perpetual-worst floor the BAU closed loop has no worst-case fixed point at all, so every trajectory declines; started from the domain ceiling of 10^4 kt it reaches 4689.6 kt after five steps and 4424.5 kt after six, which clears the six-year threshold (3787.0 kt) but not the

seven-year one (4560.3 kt). Under the informative 10th-percentile floor the loop does have an attracting worst-case fixed point (4606.5 kt for BAU, above the seven-year threshold and below the eight-year one), which is why the horizon is one year longer there. The certified sets at those horizons lie above the entire observed range of the stock. $\square$

Proposition 2.4 names the mechanism precisely: the certified horizon is the crossing at which the geometrically growing margin $K^* + r_T$ overtakes the worst-case trajectory from the domain ceiling, and it reproduces the computed horizons exactly ($T^* = 6, 6$ and 7). On this object the binding obstruction to certified intervention claims is the expansion rate itself, not the defect magnitude. This is a failure mode qualitatively different from the companion groundwater object, where the governed map is contracting and the certified horizon is defect-bound to $T \leq 3$ years. The source-year convention lengthens the horizon by one year at every class ($T = 5 \rightarrow 6$ under the two harsher floors, $T = 6 \rightarrow 7$ under the informative floor), because it replaces the registered defect magnitude ($\varepsilon = 460.0$ kt) by the source-year one (329.0 kt) while leaving $a_{\max}$ unchanged.

The horizon is not a property of the policy, and that is the second thing the crossing shows. Table 2 gives T^* against the constant catch. It is 6 years under the perpetual-worst and 5th-percentile floors and 7 under the informative one for every catch from zero to 150 kt — the moratorium included — and for every declared reactive and graded rule as well; the crossing agrees with the committed kernel computation at all 48 (rule, class) pairs the campaign tries — 30 constant catches and 18 declared rules. Only catches outside the constructive bound shorten it: 6 years under the informative floor from 180 kt and 5 under the perpetual-worst floor from 200 kt. No declared policy lengthens it. The reason is the one the formula exposes: the margin’s year-on-year increments are 379, 437, 504, 582, 671 and 773 kt, so every one of them exceeds the entire admissible catch range $[0, 91.59]$ kt — the smallest by a factor of four — and no reduction inside that range can add a certified year. Certified years cannot be bought with conservation; the horizon is set by the expansion rate at the reference point and by nothing the manager chooses.

Table 2. The certified horizon T^* by constant catch, under each declared floor class.

Constant catch (kt)	perpetual-worst	5th pct.	10th pct.
0 (moratorium)	6	6	7
5 (BAU)	6	6	7
60 (flat cap)	6	6	7
91.59 (C^*)	6	6	7
120	6	6	7
150	6	6	7
180	6	6	6
200	5	6	6
215.2 (C_{vac})	5	6	6

3.5 Stress replay and classification

The stress replay runs the closed loop from the observed 1990 SSB (861.9 kt — already below the LRP, so the replay starts outside the safe set and is uncontrolled shock accounting rather than a kernel-membership test) with the observed 1991–1995 source-year residuals. In 1991 every policy holds the stock above the LRP: flat-0 and S1 reach 953.9 kt (the S1 switch rule observes the sub-LRP 1990 stock and prescribes zero catch, so it coincides with flat-0 in that year), business-as-usual 948.9 kt, the cascade 923.9 kt, and flat-60 893.9 kt. Yet by 1992 every policy has fallen below the LRP — flat-60 to 678.9 kt, S1 to 747.8 kt, the cascade to 713.4 kt, and BAU to 797.1 kt — because the 1992 transition carries the perpetual-worst observed shock (−329.0 kt). The sharpest drawdown is under flat-60 (to 434.3 kt in 1993), while the

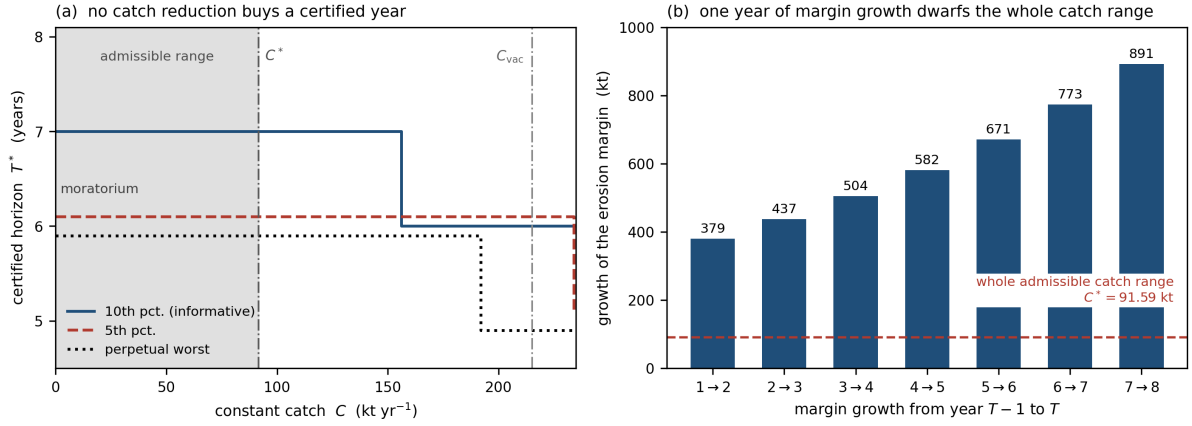


Figure 5: The certified horizon is not a property of the policy. (a) The exact crossing of Proposition 2.4 against the constant catch, under each declared floor class, at 1-kt resolution: the horizon is 6 years under the perpetual-worst and 5th-percentile floors and 7 under the informative one for every catch in the shaded admissible range, a moratorium included, and it shortens only outside the constructive bound ($C_{vac} = 215.2$ kt). Every declared reactive and graded rule lies on the same plateau. Grey band: the admissible range $[0, C^*]$. (b) Why: the erosion margin grows by 379–773 kt a year, so a single year’s growth exceeds the whole admissible catch range ($C^* = 91.59$ kt, dashed) by a factor of four at its smallest. Bars are the growth from year $T - 1$ to year T , computed from the same closed form as (a); the horizon curve reproduces all 30 archived (catch, floor) pairs exactly.

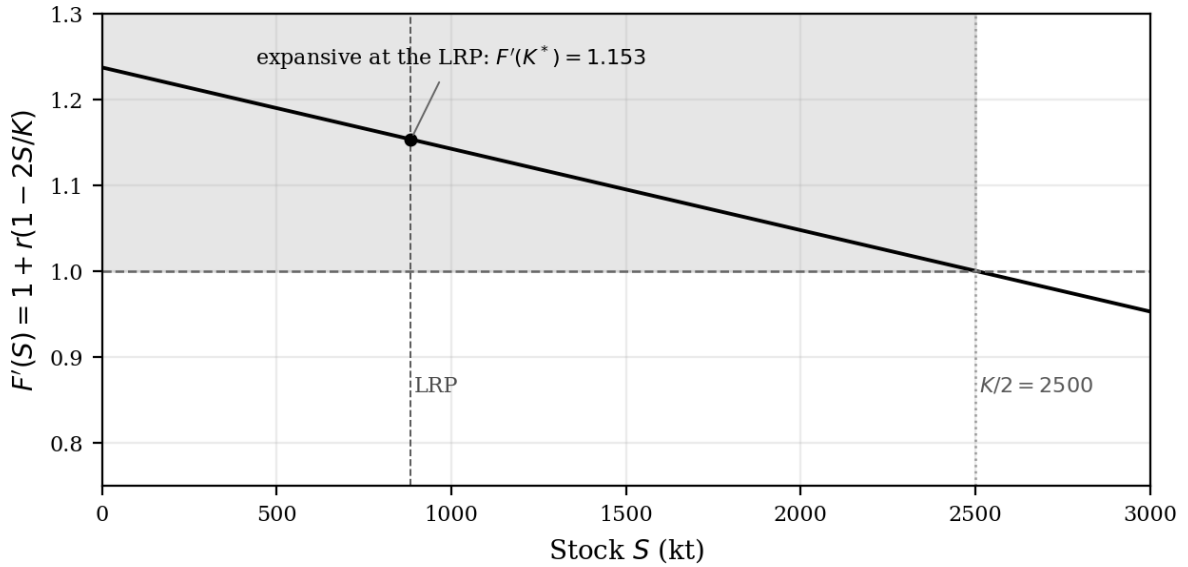


Figure 6: The closed loop’s slope $F'(S)$; the map is expansive at the LRP and contracts only above $K/2 = 2500$ kt. The expansion classification is not an artifact of the particular pinned value: the grid of Section 3.7 shows $F' \geq 1.000$ at every admissible $K \geq 2K^* = 1769.2$ kt. The classification is conditional on $K \geq 2K^*$: contraction ($F' = 0.61\text{--}0.93$) is restored only below $2K^*$ — exactly where the informative certificates of Sections 3.2–3.3 collapse and the fit cost rises — and the residual bootstrap of Section 3.10 gives F' a 90% interval of $[1.010, 1.179]$.

switch rules lose less (BAU 669.7 kt in 1994); no policy avoids the post-1992 decline. The crash is a productivity event, not a catch event — exactly the catch-insufficiency certificate of the companion forecast-evaluation study.

At the $T = 5$ classification under the 10th-percentile class, only the 1980s peak years of the 33 observed states lie inside the nominal kernels — {1985, 1987, 1989} for the 60-kt rules and additionally 1988 for BAU. The entire post-1990 history and most of the 1980s are outside. Under the perpetual-worst class all 33 are outside; under the 5th-percentile class the moratorium’s kernel is nonempty (its $T = \infty$ boundary is 2219.6 kt, above the observed range), so the classification there follows the informative geometry rather than universal emptiness.

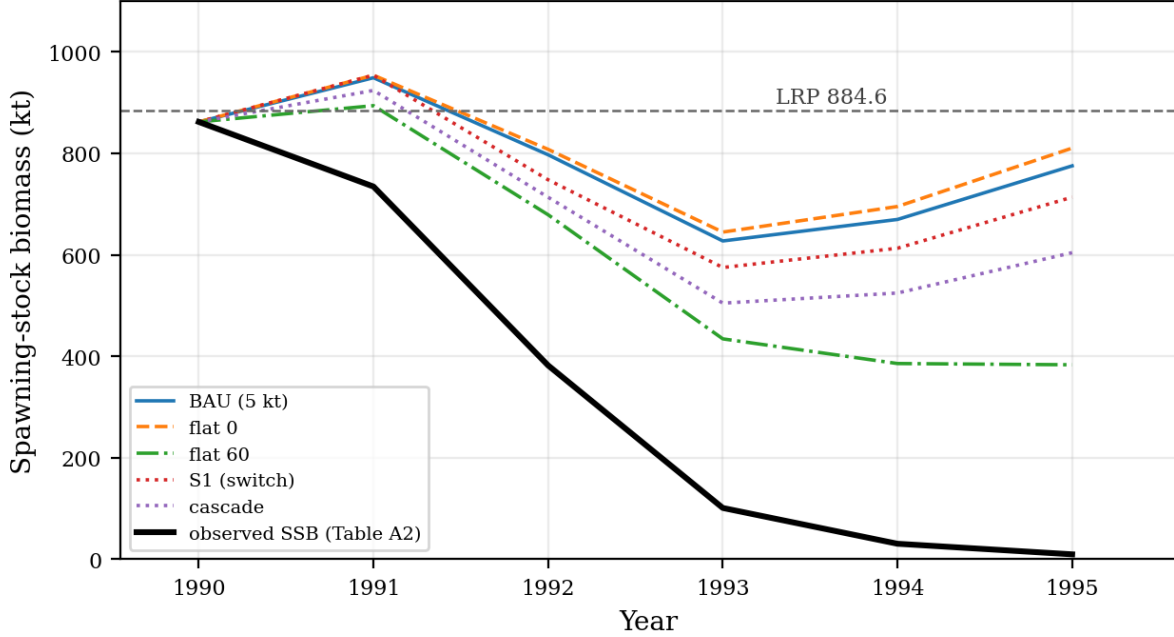


Figure 7: Closed-loop replay from the observed 1990 stock with the observed residuals; BAU, flat-0, flat-60, the S1 switch rule, and the cascade are plotted separately (the S1 rule differs from the flat-60 cap below the LRP).

3.6 Model-form comparison: the depensatory refit and the Fox form as co-equal specifications

The primary kernels are Schaefer-form (Allee term off). The depensation sensitivity refits the same object with the Allee term on, on the same 1983–2007 window with the same annual landings. The refit gives $r = 2.0$ (pinned at its optimization bound, the mirror image of the registered K), $K = 1671.7$ kt, and $s_0 = 642.3$ kt — 242 kt below the LRP, inside the observed range — with residual MSE 7690.1 kt^2 against 12,772.2 kt^2 for the registered Schaefer form. The bare fit parameters and the MSE are convention-independent (the fit already pairs the year- t catch with the $t \rightarrow t + 1$ transition), so they are unchanged from the frozen reading. Only the fitted form’s own floor classes and its kernels depend on the residual convention, and those are reported here in the source-year convention. The depensatory form is the better-fitting of the two, under identification caveats of the same kind as the registered fit.

The two forms are reported side by side as co-equal primary specifications in Table 3. The identification caveats apply symmetrically: the Schaefer form pins K at its bound, the depensatory form pins r at its bound (2.0), and neither pin is a biological finding. Table 3 reports the rows at the declared class endpoints, held frozen across forms as the corrected source-year classes (−287.4, −80.9, −329.0 kt) so that only the form varies.

Result 3.6 (Form sensitivity). The certificate directions survive the data-preferred depensatory refit. The Allee form removes the vacuity of the perpetual-worst floor — the one class vacuous on the registered form — because its maximum surplus (372.4 kt) exceeds both class magnitudes (-329.0 and -287.4 kt yr $^{-1}$).

Reason. Under the 10th-percentile class the BAU and zero-catch certificates are exactly unchanged — lower boundary 884.6 kt at every horizon, $T = \infty$ included — and the critical-zone and cascade rules are protective of the entire safe set at that class (884.6 kt). The constructive boundary of Section 3.3 and the dominance verdict of Section 3.2 are therefore unaffected — indeed the constructive bound rises, since at the declared 10th-percentile floor the Allee form gives $g(K^*) - |e_{q10}| = 196.06 - 80.87 = 115.2$ kt (123.3 kt on the declared-strength row), so the 91.59 kt certificate holds a fortiori. The clause-(H1) block itself changes shape without changing the verdict: under the Allee form the 60-kt rules are no longer empty at the 5th-percentile $T = \infty$ reading (their boundary is 1131.1 kt against BAU’s 1020.9 kt), so the block operates through a margin rather than through emptiness at that reading. Under the Allee form, whose maximum surplus is 372.4 kt, both the perpetual-worst floor (-329.0 kt) and the 5th-percentile floor (-287.4 kt) lie below it, so the BAU kernel is nonempty at those classes with infinite-horizon boundaries 1098.7 and 1020.9 kt respectively (against the empty kernels of the registered Schaefer form under the perpetual-worst floor). The expansion obstruction is not a Schaefer-form artifact: the Allee refit raises the expansion rate at the LRP from 1.1531 to 1.7818. A declared-strength alternative ($s_0 = 0.5K^* = 442.3$ kt) refits to $r = 2.0$, $K = 3223.7$ kt (MSE 9330.0 kt 2), with maximum surplus 883.6 kt; under the source-year classes its kernels are nonempty at every reported cell and lie within 12 kt (about 1%) of the identified row’s at all of them, so the depensation row does not hinge on the identification of s_0 .

Reproducibility note (two declared sensitivities). First, the manuscript’s $s_0 = 642.3$ kt is a rounding of the fitted 642.3296 kt, and one printed cell moves by a single grid step between them: the BAU worst-class $T = \infty$ boundary is 1098.75 kt at the rounded s_0 and 1098.8 kt at the fitted one (the q05 $T = \infty$ boundary moves $1020.9 \rightarrow 1020.95$ kt, which prints identically). Every other cell is unchanged, and the published cells are those of the rounded s_0 . Second, the Allee and Fox rows are computed on a 0.05 kt state grid, while the registered row comes from the committed interval-arithmetic engine; the campaign reproduces the registered row on that same grid, so the two engines can be read against each other. At finite horizons they agree to within about one grid step — of the 81 cells the campaign checks, the 72 finite-horizon ones differ by at most 0.06 kt — but adjacent to the repelling boundary, where $F' > 1$, the grid’s rounding compounds along the orbit: at $T = \infty$ five of the nine cells agree exactly and the other four differ by 0.1, 0.3, 0.6 and 0.9 kt, the largest being the BAU q05 boundary, 2218.75 against 2219.649 kt (and 2070.30 against 2070.884 kt for zero catch). The two engines are therefore not to be compared at 0.1 kt resolution, and the tolerance the campaign declares is 1.0 kt.

The third co-equal form is the Fox surplus law $g(S) = rS \ln(K/S)$, refitted one-step on the same window with the same landings and the same declared box. The refit gives $r = 0.1044$, $K = 5000$ kt pinned at its bound, residual MSE 13,873.1 kt 2 (Schaefer 12,772.2; Allee 7690.1). The Fox form is the worst-fitting of the three, and the Allee form remains the data-preferred one. The certificate directions survive the form. The expansion at the LRP is preserved ($F' = 1.0764 > 1$, weaker than the registered 1.1531). The constructive 10th-percentile bound is $g(K^*) - |e_{q10}| = 159.92 - 80.87 = 79.05$ kt, under the source-year 10th-percentile floor (-80.87 kt); the Fox form’s own refit-boundary floor is not used so that only the form varies across Table 3. The BAU and zero-catch certificates under the informative class are exactly unchanged (lower boundary 884.6 kt at every horizon, $T = \infty$ included). Three further Fox cells are computed by the campaign but not carried in Table 3: the 60-kt rule’s infinite-horizon lower boundary is empty under the 5th-percentile floor, and the 120-kt rule is empty under the 5th-percentile and perpetual-worst floors and, under the informative floor, at $T = \infty$ (its $T = 1$ boundary there is 922.7 kt). Table 3 carries the Fox row.

Table 3. Depensation and form sensitivity: kernel lower boundaries (kt) at the declared source-year class endpoints.

Form	BAU, q05 $T=1$	BAU, q05 $T=\infty$	BAU, q10 $T=\infty$	BAU, worst $T=\infty$	S1, q10 $T=\infty$
Registered Schaefer	989.0	2219.6	884.6	empty	884.6
Allee refit ($s_0 = 642.3$)	939.4	1020.9	884.6	1098.7	884.6
Declared $s_0 = 0.5K^*$ (442.3), refit	939.2	1024.1	884.6	1086.9	884.6
Fox refit ($r = 0.1044$, K pinned)	1008.4	empty	884.6	empty	884.6

3.7 Carrying-capacity sensitivity

The registered fit pins K at its optimization bound, and the expansion classification of Section 3.4 inherits the question of how much of the result is carried by that pin. The sensitivity refits r by one-step least squares at each K on the same window, in the source-year convention, with the kernel scored against each refit’s own source-year 10th-percentile floor.

Table 4. Carrying-capacity sensitivity in the source-year convention: r refit at fixed K , kernel scored against that refit’s own corrected 10th-percentile floor. Values are the BAU lower boundary of the robust viability kernel at $T = 1$ and $T = \infty$.

K (kt)	r	$g_{\max}$	$F'(K^*)$	Constr. bound (kt)	BAU q10, $T = 1$	BAU q10, $T = \infty$	q05 vacuous
1000	0.5094	127.4	0.608	−28.87	943.2	empty	yes
1200	0.5248	157.4	0.751	41.14	884.6	884.6	yes
1500	0.4099	153.7	0.926	67.9	884.6	884.6	yes
1769.2 ($= 2K^*$)	0.3559	157.4	1.000	76.55	884.6	884.6	yes
2000	0.3273	163.6	1.038	80.6	884.6	884.6	yes
2500	0.2908	181.7	1.085	85.33	884.6	884.6	yes
3000	0.2704	202.8	1.111	87.79	884.6	884.6	yes
4000	0.2485	248.5	1.139	90.31	884.6	884.6	yes
5000 (registered)	0.2369	296.1	1.153	91.59	884.6	884.6	no
7000 (out of box)	0.2248	393.5	1.168	92.89	884.6	884.6	no

Result 3.7 (Carrying-capacity grid: three readings). The grid of Table 4 supports three readings.

- (i) The expansion obstruction holds for every admissible $K \geq 2K^* = 1769.2$ kt and is the data-selected regime within that range — not an artifact of the particular pinned value $K = 5000$ — while remaining conditional on $K \geq 2K^*$: $F' \geq 1.000$ at every $K \geq 2K^*$ — exactly 1.0000 at $K = 2K^*$, rising to 1.1531 at the registered K — and the informative constructive bound rises monotonically toward the registered end of the box (−28.87 kt at $K = 1000$ kt to 91.59 kt at $K = 5000$ kt), with the fit cost rising as K falls below $2K^*$.

- (ii) Contraction is restored only below $2K^*$ ($F' = 0.61\text{--}0.93$), exactly where the informative certificates collapse: the constructive bound is negative at $K = 1000$ kt, and the BAU kernel under the 10th-percentile class is empty at $T = \infty$ with the $T = 1$ boundary raised to 943.2 kt — the moratorium itself no longer holds the reference point.
- (iii) The vacuity structure is K -dependent in the source-year convention. The perpetual-worst floor is vacuous at every in-box K , but the 5th-percentile class is vacuous only at $K \leq 4000$ kt and becomes informative at $K \geq 5000$ kt ($g_{\max} = 296.1$ kt at the registered K), which is itself the registered point.

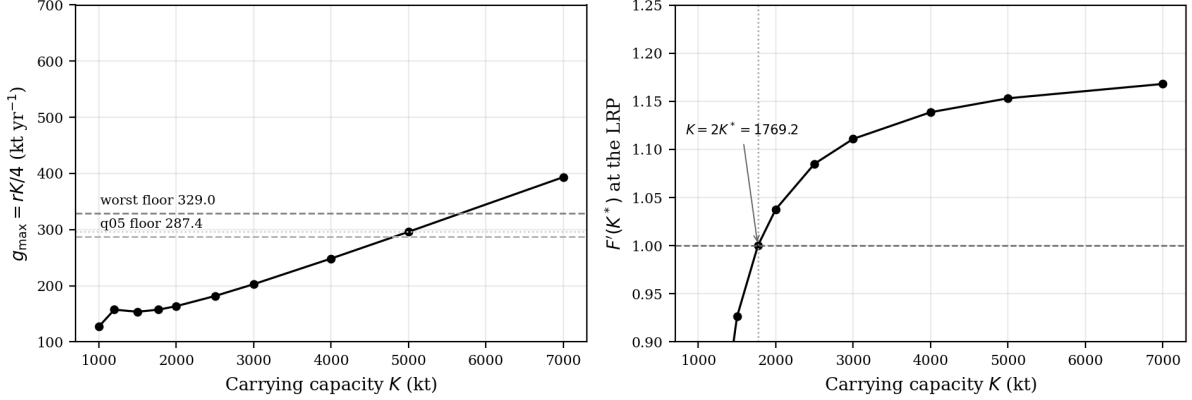


Figure 8: The two panels of the carrying-capacity sensitivity of Section 3.7.

How much of that is carried by the pin? The question can be answered without resolving it. Profiling K — refitting r in closed form at each fixed K on the same window, with the grid extended to 50,000 kt precisely because the committed value sits on the box — gives a profile-likelihood 95% set running from 1500 kt to the top of the grid: the series never approaches carrying capacity, so K is not identified from above at all. The pin itself buys little: the sum of squares at $K = 5000$ kt is 1.027 times the profiled minimum. What is identified is everything the results depend on. Across the profile set $g(K^*)$ lies in $[148.8, 176.1]$ kt and C^* in $[67.9, 95.2]$ kt, so the constructive bound is pinned to within 15% by data that do not determine K at all; and $F'(K^*) > 1$ for every profile-set K at or above $2K^* = 1769.2$ kt — the smallest expansive value on the grid is 1775 kt — which is exactly the condition reading (i) declares. The expansion obstruction is conditional on $K \geq 2K^*$; it is not conditional on the pinned value.

3.8 Stochastic viability

Persistent floors are a worst-case layer. The empirical residual pool supplies the stochastic counterpart, with the 1992 draw treated both as a member of the pool and as a removed one-off. For each policy and initial stock, 20,000 trajectories resample the 24 source-year training residuals (i.i.d., and in moving blocks of four, respecting the residual autocorrelation of 0.55). Viability is scored as the probability of remaining at or above the LRP.

Table 5. Probability of remaining at or above the LRP over 20 years ($N = 20,000$ per cell; seed fixed; draws shared across policies), source-year pool.

Policy	from the LRP, i.i.d.	from the LRP, blocks-4	from the LRP, i.i.d. without 1992	from 1500 kt, i.i.d.
zero catch	0.906	0.852	0.954	1.000
BAU (5 kt)	0.903	0.852	0.954	1.000

Policy	from the LRP, i.i.d.	from the LRP, blocks-4	from the LRP, i.i.d. without 1992	from 1500 kt, i.i.d.
60 kt / S1 / cascade	0.835	0.806	0.917	1.000
120 kt	0.647	0.650	0.769	0.999

From 1980s-peak starts (1500 kt and above) survival is near certain under every scheme — the peaks sit far above the LRP. The corrected source-year pool is marginally less severe than the frozen destination-year pool, so the survival probabilities are higher across the board.

The constructive boundary of Section 3.3 carries a stochastic reading. At the grid catch nearest the bound (92.5 kt on the campaign’s 2.5 kt grid) the 20-year survival probability from the LRP is 0.74 under i.i.d. draws, 0.73 under block resampling and 0.84 when the 1992 draw is removed — roughly three-quarters, not the near-certainty that holds at moratorium-level removals.

The $P \geq 0.9$ bar separates the two resampling schemes. Under i.i.d. resampling it is attained, but only by near-moratorium catches: the interpolated crossing is 13.5 kt against a ceiling of 0.906 at zero catch (Table 5), and by 60 kt survival has already fallen to 0.835. Under block resampling it is not attained at all, the ceiling there being 0.852 (Table 5; 0.849 in the crossing sweep, which carries its own fixed seed — a Monte-Carlo difference of 0.003 at $N = 20,000$); blocks of four preserve the residual autocorrelation of Section 2.1, so a poor year tends to be followed by another. Removing the 1992 draw lifts that crossing to 78.9 kt. The $P \geq 0.8$ crossings are 81.2 kt (i.i.d.), 72.3 kt (blocks) and 105.2 kt (no-1992). The one-off treatment of 1992 is a declared sensitivity, not an identified break.

Two properties of the pool explain the ceiling. First, failures are concentrated at the start of the horizon: the chance of an immediate breach is $2/24 = 0.083$ against a total failure probability of 0.094, and 99% of failures occur within the first three years. The reason is that from the reference point the stock is one year’s growth from safety — under zero catch a draw worse than -172.5 kt breaches the LRP at once, and two of the twenty-four residuals are worse than that, the 1992 draw (-329.0 kt) and one other (-323.5 kt). Second, a draw of that magnitude is harmless once the stock has grown past about 1020 kt, roughly a year later, which is why the ceiling is 0.906 and not the 0.43 that the $1/24$ recurrence alone would suggest: the probability of escaping the 1992 draw for twenty years is $(23/24)^{20} = 0.427$, and that would be the ceiling only if every 1992 draw were fatal.

The constructive bound of 91.59 kt sits above the 60-kt flat-cap family, so those rules are robust in the worst case, while the stochastic layer shows the survival probability falling steadily across that range (0.91 at zero-to-moratorium removals to 0.65 at 120 kt).

3.9 Finite-duration floors

A perpetual floor is a new stationary climate. The finite-duration variant asks how long a poor-productivity episode must last before the safe set contracts, with the floor active for n years and zero residual thereafter. The boundaries are computed by exact backward recursion on the monotone map, in the source-year convention (the $n = 5$ worst-floor BAU boundary reproduces the registered $T = 5$ boundary of the archived kernel table as a built-in check).

Table 6. Infinite-horizon lower boundary (kt) after a finite floor of n years followed by zero residual (source-year); empty = no state is robustly viable.

Policy	q05, $n = 5$	q05, $n = 10$	q05, $n = 15$	worst, $n = 5$	worst, $n = 10$	worst, $n = 15$
zero catch	1280.8	1510.5	1657.6	1431.6	1770.9	2015.3
BAU (5 kt)	1298.7	1540.7	1697.8	1450.0	1803.7	2062.3

Policy	q05, $n = 5$	q05, $n = 10$	q05, $n = 15$	worst, $n = 5$	worst, $n = 10$	worst, $n = 15$
60 kt / S1	1499.6	1893.2	2193.4	1656.7	2188.8	2657.5
/ cascade						
120 kt	1727.7	2329.0	2898.1	1891.7	2671.8	3562.3
180 kt	1999.0	2869.7	3986.2	2172.3	3285.8	5180.7
240 kt	2623.1	4013.4	8790.9	2825.0	4687.8	empty

Result 3.8 (Finite-duration floors). Two readings hold for the finite-duration grid.

- (i) Duration is informative. A five-year 5th-percentile episode already lifts the BAU boundary from the LRP to 1298.7 kt, and a fifteen-year episode lifts it to 1697.8 kt. The safe set contracts as the episode lengthens, without the arithmetic vacuity of the perpetual classes. The source-year floors are milder than the frozen ones, so the boundaries sit below the earlier reading at every duration.
- (ii) The ordering across policies is the same as in Table 1. Moratorium-level removals dominate every harvesting rule at every duration, and the reactive rules coincide with their 60-kt cap throughout because on the kernel domain their catch is exactly the cap's.

3.10 Uncertainty bands

The 91.59 kt constructive boundary is an analytic limit. Its sampling distribution is reported because a point would overstate the precision of the identified r . A parametric residual bootstrap ($B = 2000$, seed fixed) generates synthetic one-step series on the registered map with resampled source-year training residuals and refits r at $K = 5000$ kt on each. The refits give r a median of 0.219 (90% interval $[0.016, 0.277]$), $g(K^*)$ a median of 159.6 kt yr⁻¹ ($[11.6, 201.9]$), and the constructive bound a median of 78.7 kt with 90% interval $[0.0, 121.1]$ kt. Of these refits, 88.2% retain a positive constructive bound. The expansion classification survives the resampling: $F'(K^*)$ has median 1.142 and 90% interval $[1.010, 1.179]$. The constructive boundary is therefore best read as order 70–90 kt under the declared 10th-percentile class, not as a quota figure.

Refitting r alone at $K = 5000$ kt is the narrower of the two available bands, because it conditions the pin away rather than propagating it. Refitting both parameters on every replicate gives r a median of 0.261 (90% interval $[0.092, 0.500]$), K a median of 4191 kt with the pin active in 41.7% of replicates, and the constructive bound a median of 73.7 kt with 90% interval $[-89.4, 125.7]$ kt. The interval's negative tail is made of replicates in which the refit places K below the reference point — admissible under the declared box, and excluded by construction when K is held fixed. Restricted to the expansive regime $K \geq 2K^*$ on which the results are conditional — 72.9% of the 2000 replicates — the bound has median 88.1 kt (90% interval $[-5.6, 130.6]$ kt) and $F'(K^*)$ has median 1.137 with 90% interval $[1.030, 1.178]$: expansion holds throughout.

Two intervals for one quantity, a factor of eight apart in width, need the difference said out loud rather than left for a reader to reconcile. The profile range of Section 3.7 is an F -based interval: it holds the fit inside the 95% likelihood cut, lets K run the whole ridge from 1500 to 50,000 kt with r reprofiled at each step, and asks how far C^* can move. It answers an identification question — is the functional pinned, given the model — and the answer is yes, to within 27 kt, even though K is not identified at all. The bootstrap answers a different question: how far the estimate itself moves when the 24 transitions are resampled and both parameters are refitted freely. It is wider for two separable reasons. Its lower tail comes from replicates in which the refit places K below the reference point (7.4% of them) — admissible under the declared box, excluded by construction when K is held fixed, and not admissible as a description of a limit reference point. Removing those and every replicate outside the expansive regime still leaves $[-5.6, 130.6]$ kt, so the width is not an artefact of degenerate refits: the

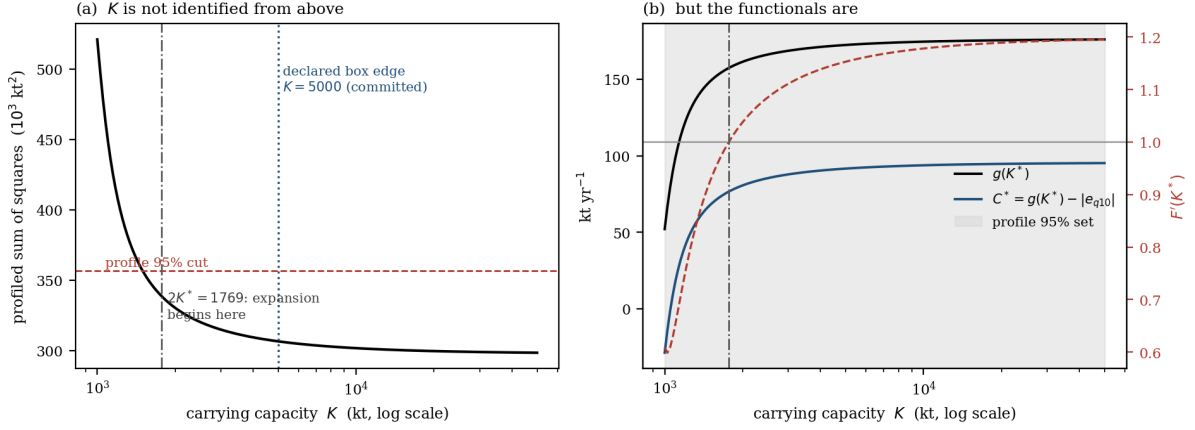


Figure 9: Identification under the pinned carrying capacity. (a) The profile sum of squares over K , refitting r in closed form at each K : the profile is flat from about 1500 kt to the top of the grid, so K is not identified from above and the committed value (dotted) buys 2.7% of the criterion. (b) The functionals the results depend on are identified nevertheless: across the profile 95% set (shaded), $g(K^*)$ and $C^* = g(K^*) - |e_{q10}|$ vary by 15%, and $F'(K^*)$ (right axis) exceeds 1 exactly above $2K^*$.

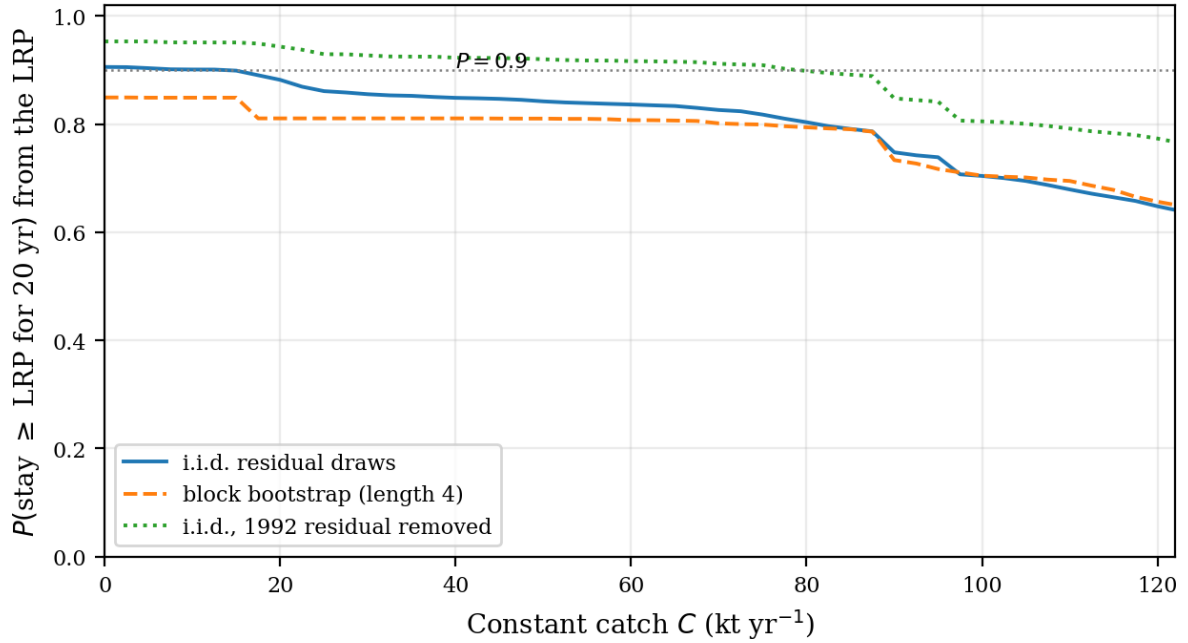


Figure 10: The three trade-off curves: 20-year survival probability from the LRP against constant catch, by resampling scheme (i.i.d. draws, blocks, no-1992).

bootstrap is wider at both ends, 35 kt above the profile ceiling and 74 kt below its floor. What remains is small-sample: with 24 transitions the resampling dispersion of a two-parameter refit exceeds what an F -based cut implies, and the profile range should not be read as a statement of precision. Neither interval is in error. They are evidence for different claims, and the claim this paper makes is the identification one.

One further declared sensitivity is made quantitative here. The residual of Section 2.1 conflates process noise and observation error, and only the process part can act as a persistent floor. If a fraction λ of the residual variance is observation error, rescaling the empirical pool about its mean by $\sqrt{1-\lambda}$ — its shape preserved, only its scale changed — moves the constructive bound from 91.6 kt at $\lambda = 0$ to 112.1 kt at $\lambda = 0.5$, and the vacuous class disappears from $\lambda \geq 0.2$ onward. The reported bound is the conservative end of that range.

3.11 Second specification (xteNCAM): a labelled sensitivity row

The registered object is the NCAM series. The second, unpooled specification is the xteNCAM series (Regular et al., 2025, Table 17; 1954–2024; LRP = 276 kt; 2024 landings persisted from 2023), refitted in the same source-year convention as the primary object (Section 2.3) on 1954–2007, with the same box rule and the safe set written against its own reference point. The floor classes are not transferred from the NCAM fit (different series scale): the row declares its own classes from its own training residuals, and no verdict transfers between the specifications. The fit gives $r = 0.5023$, $K = 4813.1$ kt (unpinned), MSE 18,028 kt², and expansion at the LRP $F' = 1.4447$ — the expansion classification is preserved and stronger than on the registered series.

The informative certificate survives under the source-year convention, but its margin is much smaller than on the registered NCAM series. For the fit’s own 10th-percentile class, $g(\text{LRP}) = 130.7$ kt and $|e_{q10}| = 123.2$ kt give a positive constructive bound of 7.49 kt. Zero catch and business-as-usual at 5 kt hold the 276-kt LRP from itself at $T = 1, 5, \infty$; their lower boundaries are 276 kt at all three horizons. Catch of 60 kt instead requires 312.45 kt at $T = 1$ and 397.60 kt at $T = \infty$; 120 kt requires 354.35 kt and 546.20 kt respectively. The harsh classes are nonvacuous ($g_{\max} = 604.4$ kt $>$ 395.9 kt); under the worst class, the $T = 1$ lower boundary runs from 462.10 kt (zero catch) to 548.05 kt (120 kt).

This labelled sensitivity therefore agrees with the registered NCAM series on expansion, catch-ordering, and self-viability at the reference point under each fit’s own 10th-percentile class. It differs sharply in margin (7.49 kt versus 91.59 kt) and in the stronger floor classes; no numeric verdict transfers between the distinct records. The 2024 xteNCAM stock (342 kt) is above that specification’s 10th-percentile zero-catch and 5-kt lower boundaries (276 kt at $T = 1, 5, \infty$), but below its 5th-percentile zero-catch one-step boundary (345.00 kt).

Table 7. The two specifications compared at their own reference points (the registered NCAM row and the second, unpooled xteNCAM row of Section 3.11).

Object	r	K (kt)	$F'(\text{LRP})$	Constr. bound (kt)	zero-catch q10, $T = 1$	zero-catch q10, $T = \infty$
NCAM (regis- tered)	0.2369	5000 (pinned)	1.1531	91.59	884.6	884.6
xteNCAM (this row)	0.5023	4813.1	1.4447	7.49	276.0	276.0

3.12 Is the reference point self-viable under current productivity?

The fit window 1983–2007 spans the 1992 collapse, so the identified r averages two productivity regimes and the floor classes average them with it. Refitting the same object on post-moratorium

windows — same source-year convention, same declared box, same construction throughout — asks whether the reference point is self-viable under recent productivity (Table 8). The rows are labelled sensitivities in the sense of Section 3.11: no verdict transfers between the two series, and none of these windows replaces the frozen protocol.

Table 8. The same construction on post-moratorium windows. K is reported with its bound status (ub: upper bound of the declared box, lb: lower bound, i.e. just above the largest observed stock on that window); C^* is the constructive bound at each window’s own 10th-percentile floor. The same quantity at each window’s perpetual-worst floor is given in the text; the residual SD, which falls from 114.9 kt on the frozen window to 18.9 kt on NCAM 1995–2015, is what drives the change.

Window	n	r	K (kt)	$g(\text{LRP})$	$ e_{q10} $	C^*	$F'(\text{LRP})$
1983–2007 (frozen)	24	0.2369	5000 (ub)	172.5	80.9	91.6	1.153
NCAM 1995–2015	20	0.2731	5000 (ub)	198.9	27.4	171.4	1.176
NCAM 1995–2007	12	0.3901	5000 (ub)	284.0	10.0	274.0	1.252
xteNCAM 1995–2024	29	0.4310	472 (lb)	49.4	41.4	8.0	0.927
xteNCAM 2005–2024	19	0.4439	472 (lb)	50.9	58.5	-7.7	0.925

On the assessment series the answer is yes, and more comfortably than on the collapse window. On 1995–2015 the fit gives $r = 0.273$, $g(\text{LRP}) = 198.9$ kt and a 10th-percentile floor of only 27.4 kt — the post-moratorium decline was smooth and therefore largely predictable one step ahead, the residual SD falling from 114.9 kt to 18.9 kt — so the constructive bound rises to 171.4 kt and stays positive under that window’s perpetual-worst floor (166.4 kt), with the expansion at the LRP intact ($F' = 1.176$). On the shorter 1995–2007 window the bound is larger again (274.0 kt).

On the second specification the answer depends on the window: the 1995–2024 10th-percentile bound is positive (8.0 kt), but on xteNCAM 2005–2024 the fit places K at its lower bound, 472 kt, barely above the largest observed stock, so the fitted surplus at that specification’s reference point is 50.9 kt against a 10th-percentile floor of 58.5 kt: the constructive bound is -7.7 kt, and on 1995–2024 it is +8.0 kt. Under either window’s perpetual-worst floor it is negative (-37.9 and -38.8 kt). The classification reverses with it — $F'(\text{LRP}) = 0.925 < 1$ — so on the modern series in its recent state the closed loop contracts at the reference point and the contraction form of the certified-layer conversion, inapplicable on the frozen object, is the applicable one.

Two caveats are inseparable from these rows and are stated rather than buried. The windows are short (12–29 transitions), and in three of the four the carrying capacity is at a bound, so these are regime readings rather than estimates of productivity. What they establish does not depend on reading them as estimates: the reference point’s self-viability is regime-dependent — comfortable on the assessment series after the moratorium, and at best marginal on the modern series, at roughly 0 ± 8 kt and negative under that series’ worst observed floor — and the expansion classification moves with it. That is consistent with Section 3.11, where the same specification’s 1954–2007 fit also failed to make its reference point self-viable.

4. Discussion

The horizon has a management consequence that is easy to miss while it is being read as a limitation of the method. A robustness certificate for a reference point has a shelf life, and Table 2 and Figure 10 show the shelf life cannot be extended by fishing less: it is six or seven years for every admissible catch, including none at all. Any interval at which the reference point is re-assessed therefore has to sit inside it, or the certificate on which the reference point’s

defence rests has already expired when it is consulted. Two real cadences bracket that. The IWC schedules an Implementation Review of its Revised Management Procedure “normally” every six years, and its aboriginal subsistence strike limits are set in six-year blocks; NOAA’s management-track assessments run on cycles of one to three years for many stocks, but for others “a decade may pass before an assessment is conducted.” A decadal cadence is outside the horizon computed here; a six-year one sits exactly at it, and only a cycle shorter than six years is comfortably inside. The same arithmetic says what would move the horizon: it is set by the expansion rate $F'(K^*)$, so neither better data about the disturbance nor a more cautious catch extends it — only a reference point low enough that the governed map contracts there, or a policy that changes the map’s shape rather than its level.

Two layers of negative content must be kept distinct. The vacuous-class identity (Section 3.2, definitional note) is a robust-layer statement: under the perpetual-worst persistent floor, no catch policy — zero catch included — holds the LRP. Under the 5th-percentile floor the statement is no longer vacuous: in the source-year convention that floor lies below the maximum surplus, and the moratorium’s $T = \infty$ kernel is nonempty (2219.6 kt). The certified-layer emptiness beyond $T = 6$ years under the two harsher floors and beyond $T = 7$ under the informative one (Section 3.4) is a different statement, about the conversion’s expansive form: the governed map’s expansion rate ($F' = 1.153$ at the LRP) empties every certified kernel at those horizons. Neither result paraphrases the other. The vacuous-class identity’s perpetual-worst reading is form-sensitive — the data-preferred compensatory refit restores a nonempty kernel at that class, the one class vacuous on the registered form (Section 3.6) — and its 5th-percentile half is not, that class being informative on both forms.

Three classes of findings must also be kept distinct — the form and specification sensitivities, the geometric findings, and the identified findings. The form and specification sensitivities carry the same division. The Fox form preserves every certificate direction at a higher fit cost (Section 3.6), and the second specification preserves the expansion classification and the policy ordering while reversing the informative certificate at its own reference point (Section 3.11). The geometric findings are form- and specification-independent; the identified findings are not; the scope is exactly as stated.

Identified findings are properties of the fitted map: the constructive boundary, the expansion rate at the LRP, and the stochastic survival probabilities of Section 3.8 all move with r and K . The boundary-harvesting rules’ “less protective by declared geometry” reading under the frozen convention is not an identified finding but a consequence of that convention’s harsh floor classes: under the corrected source-year 10th-percentile class the critical-zone rule, the cascade, and the graded rules all hold the LRP from itself at every horizon, so the geometry-based verdict of the frozen reading no longer applies. What survives is the clause-(H1) structural block at the 5th-percentile $T = \infty$ class: no positive-catch rule dominates BAU because every such rule is empty there while BAU’s kernel is nonempty (2219.6 kt). The supply comparison is a second failure, not the mechanism. The surplus-proportional family is the closest to a dominance, in that it is both reactive and protective at $\phi \leq 0.50$ under the informative class while harvesting more than the moratorium — a genuine trade-off, not a strict improvement. It fails of dominance because, on the 5th-percentile class and at $T = \infty$, it does not improve on BAU at all, and the equal-protection flat cap yields a higher mean catch. *Form-sensitive* findings reverse under the data-preferred compensatory refit, and the carrying-capacity grid of Section 3.7 shows the identified findings survive K within the declared box while the geometric ones are K -independent by construction.

The expansion obstruction is also the contribution to the viability-methods record. The certified-layer machinery used in the companion studies assumes a contracting closed loop, and the by-catch fishery case that anchors kernel computation in the fisheries literature (Krawczyk et al., 2013) is likewise a contracting setting. This cod object is the first scored instance in which the contraction form is provably inapplicable at the declared safe set — the map’s steepest

growth occurs exactly where the stock is scarcest, at the boundary the governance question is about. For a collapsed stock below half its estimated carrying capacity, every governance statement that survives certification must therefore be time-bounded and expansion-bound, not defect-bound.

Four consequences for the methods record follow from the new layers. First, the carrying-capacity grid shows the expansion classification is the data-selected regime, not an artifact of the pinned bound, so the certified-layer emptiness is a property of the identified map rather than of the conversion’s tuning. Second, the finite-duration floors separate the perpetual-class vacuity from a duration-respecting robustness question that remains informative. Third, the stochastic layer converts the worst-case certificates into survival probabilities on the empirical residual pool — the quantity a harvest-control discussion can actually weigh. Fourth, the reactive families extend the declared policy family with genuinely state-feedback rules, so the “how do reactive rules fare” question is answered directly rather than by proxy; the surplus-proportional family registers a real trade-off (protective under the informative class, more harvesting than the moratorium) while the surplus-harvest and ramp designs are dominated and recorded as rejected.

The map is one-pool surplus production on annual means. There is no age structure, migration, or survey catchability (the model-type limitations of the companion forecast-evaluation study carry over). K is pinned at its optimization bound. The expansive classification at the LRP carries the K -pin’s declared defect with it, in one reading consistent with Section 3.7 and Figure 4: $F'(K^*) = 1 + r(1 - 2K^*/K)$ exceeds 1 exactly while $K > 2K^* = 1769.2$ kt. The expansion is conditional on $K \geq 2K^*$, and within that range it is not an artifact of the particular pinned value — it holds for every admissible $K \geq 2K^*$, and the carrying-capacity grid shows the data select that range (the fit cost and the informative certificates both deteriorate below $2K^*$). Any data-supported K below twice the LRP would make the closed loop contract at the boundary and restore the contraction form of the conversion; the condition is on K , not on the identified r . The residual conflates productivity shock and model error (no observation-model separation). The closed loop observes the stock exactly at the decision instant; real governance operates under assessment lags (the one-year delay module of the companion forecast-evaluation study), so the reported kernels are upper bounds for a perfect-observation controller, and delay-aware kernels would be smaller or empty. The persistent-shock classes are deliberately harsh (a perpetual floor, not an independent draw). The 10th-percentile class is the mildest with non-vacuous content, and only the perpetual-worst floor sits beyond the map’s maximum surplus (Section 3.2), which is what makes its emptiness vacuous as a productivity statement.

The dominance rule’s protective clause is structurally conservative toward the moratorium. Under the source-year informative class the critical-zone, cascade, and graded rules match BAU’s protection exactly, and the surplus-proportional family at $\phi \leq 0.50$ holds the LRP at every horizon while harvesting more than the moratorium; but under the 5th-percentile class at $T = \infty$ every positive-catch rule is empty while BAU’s kernel is nonempty (2219.6 kt). Because clause (H1) is read at every reading with empty scored as worst, that 5th-percentile emptiness is what blocks dominance, and only a rule that is at least as protective as BAU at every reading and improves on the moratorium somewhere — none here — could dominate BAU. The Allee term is off in the primary specification; the depensation sensitivity of Section 3.6 shows the certificate directions are stable under the data-preferred Allee refit, with the expansion obstruction worsening and class-vacuity narrowing. The safe set’s upper edge (10^4 kt, twice K) is a computational grid cap, never approached on any reported kernel path; kernels are written $[s, \infty)$ because the cap is never approached, and the positive-part floor $[\cdot]_+$ never binds on any reported kernel path. Sub-LRP cascade stages are declared scenarios, not verified institutions. The certified layer is vacuous at observed stock levels.

Nothing here promotes or demotes any forecast module, transfers numbers from an interval-verified linear template (a companion methodological study, under review), or pools the ex-

tended xteNCAM series. The 91.59 kt boundary is not a quota recommendation; it is the analytic limit of robustness at one declared shock class. For management, the deliverable is the risk statement rather than the point: on this map the moratorium-level removals and zero catch are the only policies that hold the safe set under the informative class, the largest robust constant catch is of order 70–90 kt, and the stochastic layer puts a probability on holding the line (0.90–0.91 at zero-to-moratorium removals, falling with catch) that the persistent floors cannot express. The result does not transfer to Northern cod productivity or to harvest-control rules in general; it is a scored comparison on one fitted map, and the LRP’s protection in good years is the property of that map and those classes.

5. Conclusions

Scored intervention selection on the fitted Northern cod surplus-production map yields five findings and two definitional notes, stated at their actual strength.

- (1) Under the informative 10th-percentile productivity class, moratorium-level removals and zero catch hold the entire safe set at every horizon, and the largest robust constant catch is 91.59 kt yr⁻¹ — an analytic limit whose bootstrap 90% interval is [0, 121.1] kt and whose 20-year survival probability from the LRP is 0.74 under i.i.d. resampling (0.73 under block resampling, 0.84 with the 1992 draw removed), so the operational reading is order 70–90 kt, not a quota.
- (2) No non-BAU policy dominates BAU, and the mechanism is the clause-(H1) reading at the 5th-percentile $T = \infty$ class rather than boundary geometry or, alone, supply: every positive-catch rule is empty there while BAU’s kernel is nonempty (2219.6 kt). Under the informative 10th-percentile class the critical-zone, cascade, and graded rules match BAU’s protection exactly, and the surplus-proportional family at $\phi \leq 0.50$ holds the LRP at every horizon and harvests more than the moratorium — so the reactive family registers a genuine trade-off, not a strict improvement. It still does not dominate BAU: it does not improve on BAU at the harsher-informative class, and an equally protective flat 60-kt cap supplies more.
- (3) The certified layer has a finite horizon — six years under the two harsher floors, seven under the informative one — because the map is expansive at the reference point — for every admissible $K \geq 2K^* = 1769.2$ kt, so not an artifact of the particular pinned value, though conditional on $K \geq 2K^*$ — and the carrying-capacity grid shows this is the data-selected regime within that range: contraction is restored only below twice the LRP, where the informative certificates themselves collapse.
- (4) The certificates survive the data-preferred depensatory refit, and the class-vacuity reading narrows under the Allee form.
- (5) The certificate has a shelf life, and it cannot be bought. The certified horizon is six or seven years for every declared policy and every constant catch from zero to 150 kt — the moratorium included — because the erosion margin’s annual growth is 379–773 kt, four times the whole admissible catch range at its smallest. A review interval longer than that horizon is consulting an expired certificate.
- (6) The comparison is governed by an identity rather than by a tabulation: at the reference point a rule’s protection margin plus its catch equal 91.59 kt exactly (Proposition 2.1), so no reactive rule can out-supply a flat cap it matches in protection and the no-dominance verdict of finding (2) is structural. The certified horizon is likewise a crossing, not a search: six or seven years is where the geometrically growing erosion margin overtakes the worst-case trajectory.

- (7) The results are convention-dependent in the residual convention only in the sense that the source-year convention sets the constructive bound at 91.59 kt, brings the 60-kt rules onto the reference point itself (their $T = \infty$ boundary is 884.6 kt, against 900.3 kt under the registered convention), lengthens the certified horizon (to $T = 6$ under the two harsher floors and $T = 7$ under the informative one), and reduces the vacuous family to one class; every parameter and every certificate direction is otherwise stable.

Definitional note A (identity). Only the perpetual-worst floor is vacuous, and that vacuity is an arithmetic identity of the declared class — the floor exceeds the map’s maximum surplus, so every trajectory declines for every catch, zero included — not an empirical finding about Northern cod productivity (Section 3.2’s note). The 5th-percentile and 10th-percentile classes are informative, and the 5th-percentile moratorium kernel is nonempty (2219.6 kt).

Definitional note B (rule-level). The no-dominance outcome of finding (2) is a selection property of the declared partial order as much as of the policies: clause (H1), read at every reading with empty scored as worst, blocks every positive-catch rule at the 5th-percentile $T = \infty$ class given BAU’s nonempty kernel there, so within the declared family the outcome could not have gone the other way (Definition 2.6’s note). It is recorded as a note so that it is not read as an empirical selection finding on a par with the constructive bound and the stochastic layer.

The methodological content is the protocol itself — frozen scoring, layered certificates, the separation of geometric from identified findings, and the extension of the family to genuinely reactive rules — applied to a collapsed-stock reference point. The empirical content is scoped to the map, the classes, and the rule declared above. On that object the LRP is protected by good years, but the margin good years must supply is smaller than the frozen convention implied, and reactive rule design can protect the LRP but not out-supply a flat 60-kt cap. Across assessment specifications the findings are specification-conditional: on the second, unpooled xteNCAM specification (LRP 276 kt) the expansion classification and kernel ordering survive, and self-viability under its own 10th-percentile source-year class holds for zero or 5-kt catch — but its constructive margin is only 7.49 kt, not the registered series’ 91.59 kt (Section 3.11). The current stock does not meet the second specification’s 5th-percentile zero-catch one-step boundary. Across productivity regimes they are regime-conditional in the same way: the constructive bound is 171 kt on the post-moratorium assessment window and 0 ± 8 kt on the modern series (Section 3.12), so the reference point’s self-viability is a property of the regime as much as of the rule.

System II — Governance operators and viability kernels of the Edwards Aquifer

Article type: Research Paper

Abstract

Problem. Permitted groundwater systems manage drought with triggers; the protection competing rules deliver under adverse recharge is rarely scored against a fixed criterion.

Approach. We compute robust viability kernels of the J-17 annual-mean head for a policy family — training-mean pumping (282.16×10^3 acre-ft yr⁻¹), flat caps of 0–90%, a Stage-I reactive rule, and the critical-period-management cascade — under persistent drought-floor recharge. A policy is retained only if at least as protective as training-mean pumping and more permissive than the smallest securing flat cap, under a protocol frozen before any score.

Results. At the 618-ft threshold under the drought-of-record floor, training-mean pumping’s worst-case attractor (615.72 ft) lies below it, its kernel emptying beyond 13 years; every flat cut of 10% or deeper and both reactive rules make the safe set invariant; the smallest

securing cut is interpolated 7.2% (31.5% of current). The reactive rules are retained only **nominally** at 618 ft under the mildest floor class: their supply margins (+3.3% Stage I, +0.4% cascade) are hybrid, worst-case protection against wet-year replay entitlement the robust class excludes. Retention is **not** certified (positive-pumping certified kernels empty beyond $T = 3$ years), and **nothing is retained at 660 ft**, where the rules are invisible to the kernel of their own trigger — a geometric property of trigger design.

Implications. Reactive rules match flat-cap protection at the physical threshold and seem generous only if wet-year pumping counts; a positive attractor-to-threshold margin defeats every non-negative pumping rule, so the 660-ft threshold is protected by wet years, not by policy; certified claims are limited to three years by the fitted defect.

Keywords: Edwards Aquifer; critical period management; J-17 index well; pumping policies; robust viability; viability kernel; groundwater governance

Article Impact Statement. An intervention-selection protocol — governance operators, robust viability kernels under declared recharge floors, declared-defect erosion, and a frozen retention rule — is demonstrated on the Edwards Aquifer. It resolves which pumping rules protect a physical threshold under adverse recharge and at what cost in permitted supply: on J-17’s 618-ft level, reactive trigger rules match flat-cap protection at 3.3% (Stage I sketch) and 0.4% (cascade) more permitted supply than the kernel-matched flat-90% cap — nominal, hybrid-supply results — while the 660-ft institutional threshold is protected by wet years, not by the pumping family. The protocol is the generalizable content; the specific verdicts are scoped to this system.

1. Introduction

Drought management in permitted groundwater systems is built on triggers. An index-well or springflow reading crosses a threshold, and pumping reductions activate. Trigger design has a long evaluation literature: drought indicators and triggers are properly assessed as stochastic objects, with explicit attention to how indicator levels, trigger thresholds, and drought stages interact (Steinemann 2003). Robustness has been a recurring concern of water-resources planning since the recognition that optimal solutions can be brittle to the hydrologic assumptions that produce them (Watkins and McKinney 1997). The operational question for a district that already operates triggers is narrower, and comparative: given a declared family of pumping rules, which rules deliver protection under adverse recharge, and at what cost in permitted supply?

The Edwards (Balcones Fault Zone) Aquifer of south-central Texas is the most fully institutionalized example of trigger-based groundwater governance in North America. Its critical-period management program stages pumping reductions at index-well thresholds. The reductions protect springflow at Comal and San Marcos Springs, whose endangered-species requirements are administered through the Edwards Aquifer Habitat Conservation Plan, the subject of a standing scientific review by the National Academies (National Research Council 2015). The aquifer’s regional behavior has long been represented — and debated — through lumped and equivalent-porous-media models, from the Barton Springs groundwater availability model (Scanlon et al. 2003) to its recalibration by the Texas Water Development Board (Hutchison and Hill 2011). This study scores, rather than asserts, the protection supplied by a declared family of pumping rules on the measured J-17 record.

The instrument is the robust viability kernel: the set of initial heads from which a closed-loop pumping rule keeps the system inside a declared safe set under every disturbance in a declared class. Viability methods have an established lineage in natural-resource management, where constraints rather than objectives define the policy question (Krawczyk and Pharo 2013), and their computational tractability on management-scale problems is documented by fishery case studies (Krawczyk et al. 2013). The declared safe set here is a floor on the productive store rather than on a year’s yield. The head threshold marks the level below which springflow at

Comal and San Marcos approaches cessation. The policy question is therefore the protection of the base that regenerates the yield, and a trigger’s function is to keep the store from being drawn down to that floor at the cost of near-term permitted supply. The governed object is the one-pool affine head map fitted in a companion forecast-evaluation study on the same series (under separate review), which found last-value persistence unbeaten by the causal ladder (7.55-ft oracle RMSE against 13.23 ft persistence and 12.84 ft AR(1); Author et al., in review). This study asks the governance question that the forecast study could not: whether a declared pumping rule changes the viability kernel of the real system as represented by the fitted map, and at what cost in permitted supply. A companion intervention study on a marine fishery stock applies the same design and reaches the mirror verdict — none retained there, reactive rules retained here (nominally, under the mildest floor class); the mechanism contrast (fast wet-year recovery in this aquifer against a fatal productivity floor at the cod reference point) is the only reason to mention it, and the two systems’ scores are never pooled (Author et al., in review).

The complete comparison — object, defect declaration, disturbance classes, policy family, and retention rule — was frozen in a dated protocol (2026-08-26) before any score was computed.

Related work

The regulatory object. The J-17 index well is not merely a monitoring point: it is the official regulatory trigger of the Edwards Aquifer Authority’s Critical Period Management plan, and crossing its thresholds automatically imposes pumping restrictions (Edwards Aquifer Authority, 2024). Under the EAA Act, permitted withdrawals from the aquifer are capped, and the critical-period stages impose graduated withdrawal reductions on the San Antonio Pool as the J-17 level and Comal and San Marcos springflows decline; the Stage I trigger for the San Antonio Pool is a J-17 level below 660 ft MSL with a 20% withdrawal reduction, rising to a 40% reduction at Stage IV; the Uvalde Pool is staged separately on J-27 (Edwards Aquifer Recovery Implementation Program, 2021). **The 660-ft threshold studied in this paper is therefore not an arbitrary level chosen for the analysis but the actual Stage I trigger**, which is what makes the results below directly interpretable as a statement about the instrument in force. The 618-ft level is the physical threshold at which the analysis finds protection to be a different question.

Viability applied to groundwater. The viability approach has been applied to aquifer management repeatedly, and the comparison it invites is by now standard: in the optimal-control framing an environmental objective enters as a monetised externality, while in the viable framing it enters as a constraint to be satisfied, and the two produce different trajectories and different long-run levels (Pereau, Pryet and Rambonilaza, 2019). The viability kernel has been identified analytically in hydro-economic models of aquifers under conflicting food-security and ecosystem objectives (Pereau, Mouysset and Doyen, 2018), and the wider survey literature records reservoir and groundwater management as established application areas (Oubraham and Zaccour, 2018).

What this paper adds to that line is the *robust* element and the geometry of the trigger. The kernels here are computed against a persistent drought-floor recharge rather than a nominal one, and the central negative finding — that **nothing is retained at 660 ft, because the rules are invisible to the kernel of their own trigger** — is a statement about the relationship between a trigger level and the reachable set under it, not about the relative merits of optimal and viable control. That is a different question from the one that literature asks, and to our knowledge it is not asked there.

From safe yield to sustainability, which this paper must acknowledge. It is a conspicuous feature of the groundwater literature that the classical management concept — safe

yield, the balancing of annual withdrawal against annual recharge — was shown to be unsustainable in principle, precisely because it ignores the natural discharge that recharge already supports (Sophocleous, 1997; Alley, Reilly and Franke, 1999; Sophocleous, 2000; Alley and Leake, 2004). The persistence of the underlying water-budget reasoning has been documented as well (Devlin and Sophocleous, 2005). **This paper does not use the term, and should be read against that background.** Its finding that a positive attractor-to-threshold margin defeats every non-negative pumping rule — so that the 660-ft threshold is protected by wet years rather than by policy — is the robust-viability analogue of the safe-yield critique: a level that looks protected under average or favourable conditions is not protected under the adverse recharge the constraint set is required to survive.

Viability theory. That the governing question is which states admit controls keeping every constraint satisfied over time, rather than which policy optimises, is the founding move of viability theory (Aubin, 1991); the obstruction calculus underlying the certificates used here is the observation-constrained development of that programme.

2. Methods

2.1 Object and dynamics

The object is that of the companion forecast evaluation: J-17 annual-mean head H_t (ft above mean sea level), San Antonio Pool, 1934–2023, drawn from the registered twenty-column analysis panel (that evaluation’s fixed analysis dataset — the measured head, the constructed recharge and pumpage series, and the service and derived series declared there — of which only the head, recharge, and pumpage columns enter this analysis). The head is a measured well level, not an assessment inversion. Recharge R (USGS-estimated, Puente 1978 method; Umphres and Choi 2025) and pumpage P (Edwards Aquifer Authority Table 1) are the fluxes. No new data enter the analysis.

The dynamics are the causal stock-flow class of the companion forecast study, with one pool and affine form:

$$\Delta H_t = \alpha + \beta R_t + \gamma P_t + \delta H_{t-1}.$$

Definition 2.1 (One-pool affine head map). Let H_t denote the annual-mean head (ft above msl) at J-17, R_t the annual recharge, and P_t the annual pumpage. Under the hypotheses (F1) one-pool storage, (F2) affine dependence on (R_t, P_t, H_{t-1}) , and (F3) ordinary-least-squares fitting on 1934–1990 (56 transitions), the head map is the relation above.

The fit is $\alpha = 163.49$, $\beta = 0.0198$ ft per 10^3 acre-ft, $\gamma = -0.02844$ ft per 10^3 acre-ft, $\delta = -0.2539$, so $a = 1 + \delta = 0.7461$. The contraction $a = 0.7461$ corresponds to 25.4% per year mean reversion (the companion’s rolling AR(1) on the same series gives 0.66 — different window, different object; the full-sample coefficients there are (0.017, -0.026) against this window’s (0.0198, -0.02844)). Residual SD is 5.60 ft; training maximum is 15.41 ft. Out of sample (1991–2023, audit only, no refitting), the SD is 8.40 ft and the maximum is 21.81 ft. The out-of-sample maximum exceeds the declared defect. This fact is recorded below and not repaired, per protocol. The coefficients are fitted on the largely unregulated 1934–1990 era, so ordinary least squares cannot separate the aquifer’s physical response from the pumpers’ historical drought reactions, and imposing state-dependent rules under those coefficients assumes the physics-behavior separation the estimator does not guarantee (Lucas 1976); this bears on whether γ may be read as a policy lever at all, and the companion found the sign of γ reversed on the 1934–1950 subwindow.

The information pattern is the following: the manager ends year t knowing (H_t, R_t, P_t) and sets $P_{t+1} = \pi(H_t)$. The recharge R_{t+1} is unknown at decision time and is treated adversarially within a declared persistent floor. The kernels therefore assume year-end

J-17 is known exactly; the actual CPM triggers on 10-day means and cuts within the year, so annual triggering overstates the lag and understates the cut-active fraction.

2.2 Uncertainty classes and safe sets

The disturbance classes are persistent recharge floors computed on the training window: UC-min = 43.7 (the 1956 drought-of-record year, held perpetually), UC-q05 = 166.5, and UC-q10 = 179.1×10^3 acre-ft yr⁻¹. UC-q05 and UC-q10 are the 3rd and 6th lowest recorded annual recharges of the 57 training years (linear-interpolation percentile estimator); the floors themselves have historical precedent, and only their perpetuity is harsher than any recorded year. They are certification geometry, not recharge forecasts.

The safe sets are the two declared normative thresholds, fixed in the frozen protocol: $K_{\text{phys}}^* = 618$ ft (Comal Springs cessation proximity) and $K_{\text{inst}}^* = 660$ ft (the post-2007 Stage I trigger, not applied to pre-2007 history; a 10-day mean in the actual institution, an annual mean here). The model domain is [610, 710] ft. Upward exits above 710 ft are model-domain exits, not threshold violations.

2.3 Governance family

The declared governance family is the following. Training-mean pumping (called BAU in the frozen protocol; the label is retained in tables and, under this declaration, in running prose as the family’s baseline label) holds pumping at the 1934–1990 training mean, $P \equiv \bar{P} = 282.16 \times 10^3$ acre-ft yr⁻¹ — a flat 100% cap on the unregulated mean, not the historical pumpage path and not current use. Flat caps prescribe $\rho \bar{P}$ with $\rho \in \{0.9, 0.8, 0.7, 0.6, 0.5, 0.0\}$. The Stage-I reactive rule (S1) cuts pumping 20% when $H < 660$ ft; the reduction is a 20%-at-660 sketch of the Authority’s published Stage I (Edwards Aquifer Authority, Critical Period / Drought Management; the actual rule keys on 10-day J-17 or Comal flow). The critical-period-management (CPM) cascade prescribes cumulative stage totals of 20/30/35/40% cuts at $H < 660/650/640/630$ ft (cumulative, not stacked). The actual program declares five stages (Stage V at 625 ft); only the four-stage sketch is scored, Stage I is verified against the published 20%/660 pair, and stages II–IV are declared scenarios, not verified institutions.

The CPM supply figure below is the stage-weighted evaluation on the observed occupancies of the training era (1934–1990, $n = 57$ years): Stage I ($H < 660$ ft) 20 years (35.1%), Stage II ($H < 650$ ft) 12 (21.1%), Stage III ($H < 640$ ft) 5 (8.8%), and Stage IV ($H < 630$ ft) 1 (1.8%; 1956). The cumulative 20/30/35/40% cascade evaluated on these occupancies reproduces the replay’s mean prescribed pumping exactly (254.93×10^3 acre-ft yr⁻¹). Out of sample (1991–2023, $n = 33$ years) the occupancies are 33.3%, 15.2%, 6.1%, and 0.0%. Stages II and III recur out of sample (15.2%, 6.1%); Stage IV does not (0.0%; 1956 alone is below 630 ft), so CPM’s deepest cut is a 1956-only scenario in the historical measure.

Definition 2.2 (Robust viability kernel). Fix a policy π , a disturbance class $\mathcal{D}$, a safe set K^* , and a horizon T . The robust T -step viability kernel $\text{Viab}_T(\pi, K^*, \mathcal{D})$ is the set of initial heads from which the closed loop $P_{\{t+1\}} = \pi(H_t)$ keeps H inside K^* under every recharge sequence in $\mathcal{D}$.

For each policy and disturbance class, the robust T -step viability kernel of the safe set is computed by iterating the worst-case closed loop. The nominal kernel is reported without erosion. The certified layer applies the defect-to-margin conversion in the form the fitted map admits. The term “kernel” is used in its closed-loop reading — the robust positively invariant set of a fixed declared policy — not in the classical viability reading of Aubin (1991), which is existential over controls. No control choice enters the kernel computation. Only the evaluation of declared rules under the disturbance floor enters. With the declared policies all deterministic functions of the observed head, each robust kernel reduces to the invariance statement of a one-dimensional closed loop — attractor and domain-top arithmetic rather than control synthesis.

The word is retained because the object answers the same question (“from which states can the constraint be held?”) in the policy-fixed form.

2.4 Retention rule and certified layer

The retention rule decides which non-training-mean policies earn their additional structure. The certified layer converts the declared defect into an erosion margin on the kernel.

Definition 2.3 (Retention rule). A non-BAU policy is retained only if (R1) its robust kernel is at least as protective as training-mean pumping’s at every reading, compared on the kernel lower boundary with empty = worst, and (R2) it permits more pumping than the smallest securing flat cap with matched protection — on the declared 10% grid this resolves as the largest-supply member of the matched class (flat-90%); the grid-resolution sensitivity is reported in Section 3.4 (Table 4).

The scoring regimes are declared here rather than left implicit. **Protection is scored under the disturbance classes** — the perpetual floors of Section 2.2. **Supply is scored as the 1934–1990 historical replay mean.** The retention verdict is therefore a hybrid: worst-case protection at historical-mean entitlement, not a within-scenario dominance. Under the floor classes alone, the reactive rules are identically their matched flat caps (the cut is active every year) and carry no supply margin. The retention decision is made at the nominal level and re-checked at the certified level. The certified re-check holds only over the horizons where the certified kernels are nonempty — a horizon-truncated object, stated as such whenever it is invoked. At 618 ft under UC-min, R1 is close to automatic: every cut of 10% or deeper in the family makes the safe set invariant, so retention reduces to R2 (supply at matched protection).

Calculation 2.1 (Certified erosion margin). Let $\varepsilon = 15.41$ ft denote the declared uniform defect (the training maximum residual) and let $a = 0.7461$ denote the closed-loop contraction rate. Under the hypothesis (F4) that the autonomous contraction rate applies uniformly to the residual, the T-step erosion margin is

$$r_T = \varepsilon \frac{1 - a^T}{1 - a},$$

giving $r_1 = 15.41$, $r_3 = 35.49$, $r_4 = 41.89$, $r_5 = 46.66$, $r_\infty = 60.70$ ft.

Derivation. Under (F4) the closed-loop residual obeys $e_T = a^T e_0$. Summing the geometric series of perturbations of magnitude at most ε yields $r_T = \varepsilon(1 + a + \dots + a^{T-1}) = \varepsilon(1 - a^T)/(1 - a)$; the result is the triangle-inequality bound for bounded additive perturbations of a contracting scalar map. Numerical evaluation at the listed horizons gives the listed values. $\square$

Definition 2.4 (Certified kernel). The certified kernel at horizon T is the nominal kernel of $K^* + r_T$, with r_T as in Calculation 2.1.

Two properties of the certified layer must be stated before the results, because the audit correctly flagged the earlier account as imprecise. First, in this affine family every policy enters the map only through the intercept (γP is a constant shift), so the contraction rate a is identical across policies and (F4) is exact, not an approximation: the certified kernel is the closed-loop nominal kernel of a raised threshold, feedback included. Second, the mechanism by which the reactive rules collapse at the certified level is trigger-band entry: the eroded threshold $K^* + r_T$ crosses the 660-ft trigger between $T = 4$ ($618 + 41.89 = 659.89$) and $T = 5$ ($618 + 46.66 = 664.66$), so any threshold-erosion certificate makes every below-660 trigger rule identical to training-mean pumping from $T \geq 4$ — the same trigger-on-boundary invisibility as Section 3.2, re-entering by construction. The constant- r_T form is also conservative by construction: a per-step tube $K^* + r_t$ ($t = 1..T$) is tighter than $K^* + r_T$ held at every step.

3. Results

3.1 Worst-case attractors and the minimal cut

Table 1. Worst-case attractors of the closed loop (ft), by policy and recharge floor.

Policy	UC-min	UC-q05	UC-q10
training-mean pumping (BAU)	615.72	625.31	626.29
flat-90%	618.88	628.47	629.45
flat-80%	622.04	631.63	632.61
S1 (20% cut < 660 ft)	622.04	631.63	632.61
flat-70%	625.20	634.79	635.77
CPM cascade	628.36	636.37	637.35
flat-60%	628.36	637.95	638.93
flat-50%	631.52	641.11	642.09
flat-0 (zero pumping)	647.32	656.91	657.90

-case attractor of the closed loop under the perpetual drought-of-record floor (UC-min)

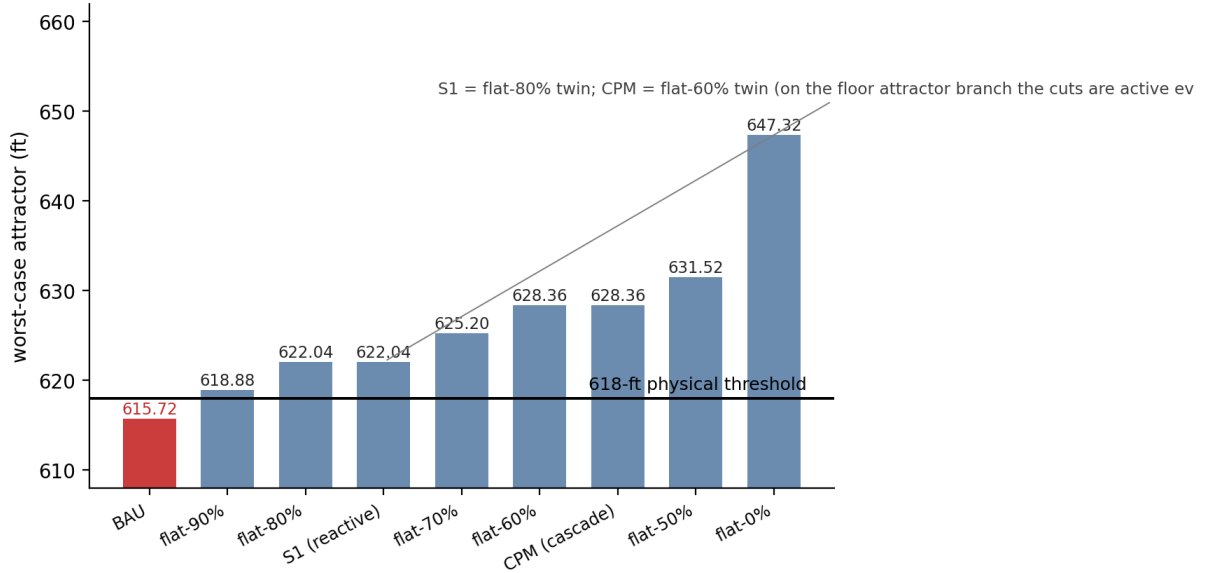


Figure 11: Worst-case attractor of the closed loop under the perpetual drought-of-record floor (UC-min) by policy, from the registered kernel computation (Table 1). Training-mean pumping’s attractor (615.72 ft) sits below the 618-ft physical threshold. Every cut policy — flat caps of 10% and deeper, the Stage I reactive rule, and the critical-period-management cascade — holds its attractor at or above the threshold. Under UC-min the reactive rules coincide with their attractor twins (S1 = flat-80%, CPM = flat-60%; under UC-q05 the CPM attractor sits between the Stage III and IV triggers and equals a ~65% cap); flat-90% clears the threshold by 0.88 ft — a margin the bootstrap of Section 3.7 shows is not resolved by the fit.

Under a perpetual 1956-recharge floor, training-mean pumping’s attractor (615.72 ft) sits below the physical threshold. The smallest flat cut whose attractor clears 618 ft is an interpolated **7.2%** of training-mean pumping, which lies outside the declared family (on the current-mean baseline of 382.16×10^3 acre-ft yr⁻¹ the same arithmetic gives a securing cut of about 31.5%, and the Stage-I 20% cut does not secure — Section 3.4). The Stage-I reactive rule’s attractor equals flat-80%’s, because the cut is active on the entire attractor branch.

3.2 Nominal kernels

Table 2. Nominal kernel lower boundary (ft) at the 618-ft threshold under UC-min, by horizon. Cut policies (flat-90% and deeper, S1, CPM) hold the whole safe set [618, 710] at every horizon, including the infinite horizon.

Horizon T	1	5	10	12	12.7 (crossover)	∞
BAU boundary	618.8	625.6	658.4	692.6	710	empty

At the physical threshold (618 ft) under UC-min, training-mean pumping’s kernel boundary climbs 618.8 ($T = 1$) $\rightarrow$ 625.6 ($T = 5$) $\rightarrow$ 658.4 ($T = 10$), and the kernel is empty beyond $T \approx 13$ years. The emptiness is a domain-top event, and its arithmetic should be read exactly: the boundary $B(T) = H^* + (618 - H^*)/a^T$ reaches the declared safe-set ceiling when the *required initial head* exceeds 710 ft — trajectories fall toward the attractor, they do not exit upward. The horizon is a function of the arbitrary ceiling: $T_{\text{empty}}(C) = \ln((C - H^*)/(K^* - H^*))/\ln(1/a)$, so the 710-ft ceiling gives 12.7 years, the observed maximum annual mean (691.96 ft, the 1992 mean) gives 12.0 years, and with no ceiling the kernel is never empty in the domain sense — every state converges to the attractor, and “empty beyond T” means “no initial head in the domain lasts T years.” With the ceiling stated, business-as-usual is not robustly viable against a perpetual drought-of-record, and the observed 1992 mean (691.96 ft) sits almost at the $T = 12$ boundary. Every cut policy in the family — a 10% flat cap, S1, CPM, and deeper — makes [618, 710] robustly invariant: the whole declared safe set is the kernel at every horizon, including the infinite horizon. Under UC-q05/UC-q10 the safe set is already invariant at training-mean pumping (attractors 625.3/626.3 ft); governance differentiates the kernel only under the drought-floor class.

At the institutional threshold (660 ft) the result is a **design observation about scoring a trigger against its own level**, and the kernel statement splits by policy class. For the **reactive rules** (S1, CPM) the equality with training-mean pumping is exact at every horizon: the boundaries (675.1 at $T = 1$, 695.3 at $T = 2$ under UC-min; empty from $T = 3$) lie strictly above the first-stage trigger (660 ft, the highest of the four declared trigger levels), no reactive rule fires in the viable region, and the kernel is policy-invariant for them. That equality is a geometric identity — a trigger placed on the boundary of a constraint is invisible to the viability kernel of that same constraint, and no rule that cuts only below 660 ft can improve the invariance of $\{H \geq 660\}$ — and it is a category observation about the scoring, not a deficiency of the rules: the cascade’s institutional purpose is springflow protection near 618 ft, not head invariance at 660 ft. For the **flat caps** the equality is false. They move the finite-horizon boundaries (flat-80% reads ≈ 673 at $T = 1$ against BAU’s 675.1) and extend the viable horizon (from $T = 3$ at BAU through $T \approx 6$ at flat-50% to $T \approx 6$ –11 for zero pumping by the same ceiling arithmetic; zero pumping’s attractor 647.3/657.9 ft still sits below 660 ft). Demand management therefore extends the viable **horizon, not invariance**, and the correct statement of the 660-ft reading is: no policy in the declared family, zero pumping included, holds $\{H \geq 660\}$ invariant under the drought floors — the threshold is protected by wet years, not by the pumping family. This is the frequency-management rationale the actual CPM rule implements, and that rationale is outside the robust-kernel frame.

The negative reading is in fact stronger than “the declared family” suggests, and the stronger form is worth stating because it is what makes the result portable. Write the fitted map as $H_{t+1} = aH_t + \alpha + \beta R_t + \gamma P_t$ with $0 < a < 1$ and $\gamma < 0$ (here $a = 1 + \delta = 0.7461$ and $\gamma = -0.02844$, from the Section 2.2 fit; both are fitted signs, reported rather than assumed). Because $\gamma < 0$, any non-negative pumping sequence $P_t \geq 0$ gives $H_{t+1} \leq aH_t + \alpha + \beta F$ pointwise under a recharge floor F ; because $0 < a < 1$, that comparison map contracts to the zero-pumping attractor $H_0^* = (\alpha + \beta F)/(1 - a)$. So whenever the *attractor-to-threshold margin* $m = K - H_0^*$

is positive, every non-negative pumping rule — not merely every member of the declared family — fails to hold $\{H \geq K\}$ invariant. At 660 ft the margin is positive under all three floors, which is why zero pumping already fails there.

Two consequences follow. First, the argument quantifies over pumping *sequences*, so it is indifferent to a rule’s functional form: it covers rules that respond to several state variables at once, and rules with hysteresis (a stage entered on one indicator and exited only on another), neither of which the declared family here can express. A district operating a richer rule set than the one scored here therefore inherits the same negative verdict. Second, the converse does not hold. A negative margin says only that zero pumping *could* hold K ; whether some member of a particular declared family does so is a separate, family-relative question, and one that a larger family can answer differently. Emptiness generalizes; construction does not.

3.3 Certified kernels

With the erosion of Section 2.4 applied, every positive-pumping policy in the family has an empty certified kernel beyond $\mathbf{T} = \mathbf{3}$ years at the physical threshold and beyond $T = 1$ year at the institutional threshold. Zero pumping is the exception. Its certified physical-threshold kernel is nonempty through $T = 4$ under UC-min (lower boundary ≈ 687.9 ft from the certified-kernel algebra; the registered horizon grid $\{1, 2, 3, 5, 8, 10, 15, 20, \infty\}$ does not sample $T = 4$ — the algebra is the record there and the grid is annotated accordingly) and through $T = 5$ under UC-q05/UC-q10. The certified boundaries at $T = 3$, UC-min, 618 ft: flat-0 662.2 < flat-80 697.8 < BAU = S1 = CPM 706.7 ft. The certified-level inference is drawn explicitly: at the one certified horizon tabulated, the reactive rules inherit training-mean pumping’s boundary (706.7) — strictly above flat-80%’s (697.8) — so certified dominance of S1 over its matched flat cap fails. The +3.3%/+16.2% supply margins are nominal-level comparisons. Certified retention is a different, horizon-truncated object from nominal retention (Section 2.4).

The bound is optimistic, and the audit’s sensitivity is recorded: at the out-of-sample defect ($\varepsilon = 21.81$ ft) the erosion margins scale to $r_3 = 50.22$, $r_5 = 66.04$, $r_\infty = 85.90$ ft, so certified emptiness beyond $T = 3$ is a *lower* bound on how fast certification dies, and certified horizons do not appear as findings beyond this section’s scope statement. The binding constraint on certified intervention claims is the model defect, not the governance. The certified comparison is also conservative in form against the state-dependent rules for the geometric reason of Section 2.4 — the eroded threshold enters the trigger band from $T = 4$ — which is a property of threshold-erosion certificates generally, not of these rules.

3.4 Supply and retention

Table 3. Mean prescribed pumping: the frozen open-loop replay (actual observed heads, 1934–1990) and the post-freeze closed-loop replay (head simulated by the map under actual recharge, same policy in force), 10^3 acre-ft yr^{-1} .

Policy	Open-loop supply (frozen)	Cut active	Closed-loop supply (post-freeze)	Closed-loop cut active
BAU (training-mean)	282.16	0%	282.16	0%
flat-90%	253.94	100%	253.94	100%
flat-80%	225.73	100%	225.73	100%
flat-70%	197.51	100%	197.51	100%
S1	262.36	35.1%	258.98	41.1%
CPM	254.93	35.1%	254.70	41.1% (any stage)
flat-60%	169.29	100%	169.30	100%
flat-50%	141.08	100%	141.08	100%
flat-0%	0.00	100%	0.00	100%

Out-of-sample open-loop replay (audit only, over the 1991→2022 transitions): S1 264.5 and CPM 260.6×10^3 acre-ft yr⁻¹; closed-loop from the observed 1990 head, S1 prescribes 268.5 with the cut active 24.2% of years and ends 2023 at 643.3 ft (observed 635.7; training-mean pumping ends 639.2). Every flat policy prescribes its cap throughout, so its supplies coincide by construction. The closed loop is a model object: run from the observed 1934 head it ends 1990 at 677.6 ft against the observed 645.8 ft (+31.8 ft), so the closed-loop supplies carry the map’s own level bias and are reported as the audit requested — as the within-model check on the hybrid margin, not as a second retention leg.

Table 4. The retention margins under every comparator reading (open-loop supplies; the closed-loop margins follow in the text).

Comparator (matched protection at 618 ft / UC-min)	Supply	S1 margin	CPM margin
kernel-matched	253.94	+3.3%	+0.4%
flat-90% (declared grid)			
attractor twin	225.73 / 169.29	+16.2%	+50.6%
flat-80% (S1) / flat-60% (CPM)			
interpolated 7.2% cut (outside family)	261.84	+0.2%	−2.6% (fails)
flat-92% (1% grid)	259.59	+1.1%	−1.8% (fails)
flat-93% (1% grid)	262.41	−0.02% (fails)	−2.9% (fails)

- **S1: retained (nominal, UC-min, 618 ft; hybrid).** It matches the flat caps’ robust invariance (kernel = whole safe set at all horizons) while supplying 262.36 versus flat-90%’s 253.94 ($+8.4 \times 10^3$ acre-ft yr⁻¹, +3.3%) and flat-80%’s 225.73 (+36.6, +16.2%). The scoring regimes are the hybrid declared in Section 2.4 — protection under the perpetual worst-year recharge floor, supply as the 1934–1990 historical replay mean. The reading of the retention is exactly “worst-case protection at historical-mean entitlement,” not a within-scenario dominance. Under the floor itself S1 is identically flat-80% (the cut is active every year, and its floor-attractor supply equals flat-80%’s), so the supply margin exists only in wet years that the robust class excludes. Against flat-90% the floor comparison is the sharper one — S1 is always −20% where flat-90% is always −10%, at identical invariance. The reactive architecture justifies its additional structure on the hybrid criterion, not on the robust criterion alone, and the comparator table shows how grid-dependent the margin is: against the 1%-grid smallest securing cap it is +1.1%, and against the interpolated 7.2% cut it is +0.2%.
- **CPM: retained (nominal, same threshold and class).** Attractor 628.36 ft (equal to flat-60%’s) at supply 254.93 versus flat-60%’s 169.29 (+50.6%) and flat-90%’s 253.94 (+0.4%). The +50.6% is priced substantially off Stage IV, a stage occupied in 1 of 90 years.
- **Closed-loop re-check (post-freeze layer).** With each policy in force and the head simulated by the map under actual recharge, S1’s supply falls to 258.98 (−1.3% versus the open-loop replay; the cut is active 41.1% of years, not 35.1%, because the map’s trajectory dips below 660 ft more often than the observed record did) and CPM’s to 254.70 (−0.1%). The hybrid margins attenuate but survive: S1 +2.0% versus flat-90% (+14.7% versus flat-80%), CPM +50.4% versus flat-60% (+0.3% versus flat-90%). The margin is a real property of the rule’s wet-year behavior on this map, not an artefact of evaluating the rules on the observed (unregulated-era) heads alone.

- **Certified level:** certified retention is horizon-truncated. At the one tabulated certified horizon the reactive rules inherit training-mean pumping’s boundary (706.7 ft, Section 3.3) and fail the certified re-check. The supply figures are nominal-level comparisons, not certified retention.
- **Institutional threshold: nothing retained.** The reactive rules are identically training-mean pumping at every horizon (trigger-on-boundary invisibility, Section 3.2). The flat caps are more protective at finite horizons and extend the viable horizon. No policy meets the retention test there.
- **Current-baseline sensitivity (post-freeze, analytic).** The declared family and every supply figure are relative to the 1934–1990 training mean (282.16). On the 1991–2023 mean actual pumpage (382.16×10^3 acre-ft yr⁻¹), the same attractor arithmetic gives training-mean-pumping 604.52 ft, the securing cut $\approx 31.5\%$, and the Stage-I/flat-80% attractor 613.08 ft — below 618 ft, so the 20% sketch does not secure the physical threshold at current use. Re-expression of the family on the current baseline is a registered follow-up, not a scored object here.

3.5 Classification and stress replays

The $T = 5$ nominal kernel (UC-min, 618 ft) classifies the actual record. Training-mean pumping excludes exactly one actual year from its viable set — 1956, the drought-of-record year (623.15 ft annual mean, above 618 ft and above the 615.72-ft attractor: 1956 is excluded as a *starting* head whose 5-year forward path under perpetual 1956 recharge falls through the threshold, not because the 1956 mean was below 618; its daily minimum, 612.51 ft, was). S1 and CPM exclude none: the entire 90-year actual record is robustly 5-year viable under the cut rules. The $T = 5$ certified kernels are empty, so no actual year is certified 5-year viable under any policy. This is the boundary of the certified analysis.

A 1950s open-loop diagnostic (model driven by actual R, P versus actual heads, 1951–1956) records the map’s drought bias. The affine map under-predicts the decline: model 659.5 $\rightarrow$ 631.3 versus actual 659.5 $\rightarrow$ 623.2 ft, maximum error 8.1 ft, biased high. The model-based policy replays from the observed 1950 head keep all policies above 618 ft (BAU minimum 629.7, S1 634.9, CPM 637.1 ft) — note that this replay holds pumping at the training mean (282.16) while actual 1956 pumping was $\approx 321 \times 10^3$ acre-ft, so it is not the decade’s actual policy either. The open-loop bias means the true margins are smaller than the replay suggests. This is recorded, and no correction is applied.

3.6 Finite-duration recharge floors and floor-class supply (post-freeze layers)

Two declared post-freeze layers extend the persistent-floor object. The first asks how long a drought-class episode must last before the safe set contracts. The class floor holds for n years, recharge then returns to its training mean, and the infinite-horizon lower boundary at the 618-ft threshold is pulled back by exact backward recursion through the n floor years. The second computes the floor-class supply — the closed loop from the observed 1934 head with recharge held at each class floor — supplying the declared-scoring half that the historical replay cannot.

Table 5. Finite-duration floors (post-freeze): infinite-horizon lower boundary (ft) at the 618-ft threshold after n years at the class floor followed by training-mean recharge. The BAU row at $n = 5$ and $n = 10$ reproduces the registered $T = 5$ and $T = 10$ nominal boundaries (625.6 and 658.4 ft). The $n = 15$ emptiness reproduces the registered 13-year horizon bound.

Policy	UC_min, $n=5$	UC_min, $n=10$	UC_min, $n=15$	UC_q05/q10, any n
BAU	625.6	658.4	empty	618.0

Policy	UC_min, $n=5$	UC_min, $n=10$	UC_min, $n=15$	UC_q05/q10, any n
flat-90% and deeper; S1; CPM	618.0	618.0	618.0	618.0

Duration differentiates exactly one policy. Under the drought-class floor, training-mean pumping is the only policy whose boundary moves with episode length. Every cut policy holds the whole safe set at every duration, because the cuts' floor attractors sit at or above the threshold (618.9–647.3 ft, Section 3.1, Table 1). A finite drought episode followed by normal recharge pulls nothing back beyond 618 ft. Under the two milder classes the safe set is already invariant at BAU (Section 3.2), and duration is inert for every policy.

Table 6. Floor-class supply (post-freeze): closed loop from the observed 1934 head (670.4 ft), recharge held at the class floor, mean prescribed pumping (10^3 acre-ft yr⁻¹) over the training span (1934–1990) and the full span (1934–2023); end head at 2023.

Policy	UC_min supply (train/full)	UC_min end head	UC_q05 supply (train)	UC_q10 supply (train)
BAU	282.16 / 282.16	615.7	282.16	282.16
flat-90%	253.94 / 253.94	618.9	253.94	253.94
flat-80%	225.73 / 225.73	622.0	225.73	225.73
S1	226.72 / 226.36	622.0	226.72	226.72
flat-60%	169.30 / 169.30	628.4	169.30	169.30
CPM	174.49 / 172.59	628.4	187.36	187.61
flat-0	0 / 0	647.3	0	0

The table quantifies the scoring-regime reading of Section 2.4. Under the floors the reactive rules converge to their matched flat-cap attractors (S1 to flat-80%'s 622.04 ft, CPM to flat-60%'s 628.36 ft — the cut active every year on the attractor branch). Their closed-loop span-mean supply exceeds the matched caps only by the trigger-lag margin of the descent from the observed 1934 head — S1 +0.4% over flat-80% (226.7 versus 225.7×10^3 acre-ft yr⁻¹, a floor-class trigger lag), CPM +3.1% over flat-60% (174.5 versus 169.3). These are different objects from the hybrid margins of Table 4: S1's +0.4% here is the within-floor trigger lag, CPM's +0.4% there is the historical-replay margin against flat-90% — same number, different quantities, and neither is a robust-supply statement. The hybrid criterion's margins exist only in the wet years that the robust classes exclude; the floor-class supply leg is the margin's own measure of how little it would buy inside the class.

3.7 Attractor uncertainty (post-freeze layer)

Table 7. Residual bootstrap of the 1934–1990 fit (5,000 replications, seeded; no replication was non-contractive): worst-case attractor under UC-min, by policy.

Policy	Point attractor (ft)	Bootstrap mean (ft)	Bootstrap SD (ft)	5th–95th percentile (ft)	P(attractor $\geq$ 618 ft)
BAU (training-mean)	615.72	613.72	13.09	590.0 – 630.6	0.41
flat-90%	618.88	617.00	12.46	594.5 – 633.0	0.53
flat-80% / S1	622.04	620.27	11.87	598.7 – 635.7	0.65

Policy	Point attractor (ft)	Bootstrap mean (ft)	Bootstrap SD (ft)	5th–95th percentile (ft)	P(attractor $\geq$ 618 ft)
flat-70%	625.20	623.55	11.35	603.1 – 638.3	0.75
flat-60% / CPM	628.36	626.83	10.89	607.3 – 641.2	0.83
flat-50%	631.52	630.11	10.51	611.3 – 644.2	0.89
flat-0	647.32	646.49	10.03	629.0 – 661.2	0.99

The audit’s resolvability question has a quantitative answer. The *ordering* of the attractors is resolved: the flat-90%-minus-BAU gap has a 90% bootstrap band of [1.9, 5.0] ft, excluding zero. The *threshold clearance* is not: whether training-mean pumping fails 618 ft ($P = 0.41$) and whether the marginal securing caps (flat-90%, $P = 0.53$; S1/flat-80%, $P = 0.65$) clear it are coin flips at this fit; only cuts of 60% and deeper reach probabilities above 0.8, and zero pumping reaches 0.99. The nominal-level statement that a 10% cut secures the threshold rests on a 2.3-ft gap (615.72 ft to the 618-ft threshold; the marginal flat-90% attractor sits at 618.88 ft) on a map with 5.6-ft residual SD and an acknowledged 8.1-ft high bias in the drought it is scored on — the fitted map cannot resolve whether the marginal members of the family protect 618 ft. The deep cuts and the failure of training-mean pumping are the resolved parts of the result.

4. Discussion

The complete evaluation loop — measured state, calibrated stock-flow map, declared governance operators, declared uncertainty classes, viability kernels with explicit declared-defect erosion, a held-out defect audit, and a fixed retention rule — yields a **nominal selection at the physical threshold under the mildest floor class**. The reactive architecture matches flat-cap protection at 3.3% (S1) and 0.4% (cascade) more permitted supply than the kernel-matched flat-90% cap — 16.2% and 50.6% against their attractor twins, +1.1% and –1.8% against the 1%-grid smallest securing cap — under a hybrid criterion whose wet-year entitlement the closed-loop re-check attenuates but does not remove. An interpolated 7.2% mean cut (outside the declared family; about 31.5% of current use) secures the physical threshold against a perpetual drought-of-record where training-mean pumping fails within roughly 13 years (a ceiling-relative statement, Section 3.2). Two boundaries are equally part of the result: the institutional threshold is not demand-manageable to invariance under the declared classes — and, by the margin argument of Section 3.2, not under any non-negative pumping rule whatever, whatever its functional form — and the certified content is defect-bound to $T \leq 3$ years.

The reactive result is system-dependent, not architectural. Its mechanism is the aquifer’s physics. High transmissivity and rapid karst recharge make wet-year recovery fast, so a flat cap permanently taxes recoveries that a reactive rule harvests — the supply margin exists only in the wet years that the robust classes exclude (Section 3.4). A companion intervention study on Northern cod applies the identical design and retains nothing: there the reactive rules cut catch exactly where the moratorium already protects. The framework’s deliverable is the scored comparison itself; which governance architecture earns its complexity is a property of the system, and the two scored systems occupy opposite ends of that design space without bounding it.

Two layers of results must be kept distinct. The robust-kernel findings are statements about the declared pumping family under the declared floor classes. The certified-kernel findings are a different layer, produced by applying the erosion conversion to the declared training defect. With

$\varepsilon = 15.41$ ft — exceeded out of sample at 21.81 ft — every positive-pumping policy’s certified kernel is empty beyond $T = 3$ years, and the bound is optimistic rather than conservative on this object (Section 3.3). The certified emptiness is a bound on what the fitted map can certify, not a robustness statement about the physical system. The mechanism of the reactive rules’ certified collapse is geometric — the eroded threshold enters the trigger band (Section 2.4) — and the certified comparison is conservative in form against state-dependent rules for that reason.

The 660-ft design observation also reads correctly against the institution it describes. The actual CPM rule manages **frequency** — the fraction of time the head spends in critical stages — not robust invariance. The finding that no pumping rule can hold the institutional threshold under a perpetual drought floor is consistent with that design intent: the rule cannot make wet years, it prices dry ones. The National Academies’ review of the Habitat Conservation Plan (National Research Council 2015) was likewise organized around monitoring and modeling adequacy under drought. The horizon-not-invariance distinction is the robust-kernel expression of the same boundary.

The karst honesty of the object is inherited from its forecast-study companion and from the modeling lineage it descends from. The one-pool affine map is the simplest member of the lumped family whose regional adequacy was tested on the Barton Springs segment (Scanlon et al. 2003; Hutchison and Hill 2011). Conduits, the Uvalde–San Antonio divide, and unconfined recharge-zone storage remain in the residual. San Antonio and Uvalde recharge and pumpage are lumped. The actual CPM triggers are 10-day averages, so the annual-mean rule is a coarse relative of the real institution. Stages II–IV are declared scenarios, not verified institutions. Nominal kernels carry no defect margin, and the certified kernels use a training defect that the out-of-sample audit exceeds. The counterfactual swap from the largely unregulated fitting era to state-dependent rules carries the declared econometric boundary of Section 2.1 (Lucas 1976). The UC floors are certification geometry, not forecasts. The 1950s replay is biased high by 8.1 ft. Nothing in this leg promotes or demotes any forecast module, and no two-pool, karst, or solute claim is made.

Cross-system reading. The protocol is the deliverable, and its value is shown by contrast rather than by a single result. The identical intervention-selection design applied to Northern cod (Abaee 2026) retains nothing — there the reactive rules cut catch exactly where the moratorium already protects, so no policy earns its complexity. Here, on the Edwards Aquifer, the reactive architecture earns a *nominal* supply margin at the physical threshold (+3.3% Stage I, +0.4% cascade) that is real but wet-year-contingent. The two scored systems therefore occupy opposite ends of the design space — one where reactive complexity is unrewarded, one where it is marginally rewarded — and the framework does not prescribe which end a new system lands on; it shows how to find out. That is the general, transferable content: the method scores protection against the robust floor class and separates the geometric (kernel/ threshold) from the identified (defect) findings, independent of the specific system.

Practical reading. For a district that already operates triggers, the result gives two usable bounds. First, at the physical threshold, deep cuts are robust and the smallest securing cut is interpolated at 7.2% of training-mean pumping (about 31.5% of current use) against a perpetual drought-of-record — but a cap’s nominal supply advantage over a reactive rule is not robust under the declared floor classes, so a manager should not credit wet-year replay supply when selecting among equally protective rules. Second, at the institutional (660-ft) threshold nothing in the pumping family is protective: the line is priced by frequency, not held by invariance, so a manager should not expect the pumping family to substitute for wet years when the cap is evaluated there. Where certified claims are wanted (a formal finite-horizon guarantee), the statement is limited to $T \leq 3$ years by the fitted defect (train $\varepsilon = 15.4$ ft, exceeded out of sample at 21.8 ft), so any longer certified claim should be read as a bound, not a certificate.

Generalizability boundary. The protocol generalizes; the verdicts do not. Every numerical

certificate is scoped to the J-17 annual-mean one-pool affine map and the declared governance operators and floor classes, and no transfer to the Uvalde Pool (J-27) is claimed, consistent with the programme’s no-pooling discipline. Conduits, the Uvalde–San Antonio divide, and unconfined recharge-zone storage remain in the residual, and the annual-mean trigger is a coarse relative of the real 10-day rule.

Two senses of generality should be kept apart here, because the margin argument of Section 3.2 is general in one of them and not the other. Within *this* calibrated map, the negative result quantifies over *rules*: a positive margin defeats every non-negative pumping sequence, so no richer policy vocabulary — multi-variable triggers, hysteresis, finer stages — can recover invariance at 660 ft. That is a theorem about the map, and it holds whatever family a manager declares. It says nothing about *other aquifers*: whether any particular pool has a positive margin at its own threshold depends on that pool’s fitted $(a, \alpha, \beta, \gamma)$ and its own floor, which must be estimated there and reported with the hypotheses $0 < a < 1$ and $\gamma < 0$ checked rather than assumed. Generality across rules is established; generality across pools is not claimed.

5. Conclusions

On the measured J-17 record, three verdicts, not one. (1) **Nominal, physical threshold, mildest floor class:** reactive trigger rules match flat-cap robust protection of the 618-ft threshold at 3.3% (Stage-I sketch) and 0.4% (cascade) more permitted supply than the kernel-matched flat-90% cap — 16.2% and 50.6% against their attractor twins — under a hybrid criterion (worst-case protection, historical-replay entitlement) whose wet-year margin survives a closed-loop re-check attenuated (S1 +2.0% versus flat-90%). An interpolated 7.2% mean cut of training-mean pumping (about 31.5% of current use) secures the threshold against a perpetual drought-of-record, where training-mean pumping fails within roughly 13 years (ceiling-relative). (2) **Certified:** nothing — every positive-pumping certified kernel is empty beyond three years, an optimistic bound on a defect the out-of-sample audit exceeds. (3) **Institutional threshold:** nothing — the 660-ft line is protected by wet years rather than by the pumping family, a geometric property of scoring triggers against their own level. The fitted map resolves the deep-cut protections and the failure of training-mean pumping, and cannot resolve the marginal caps ($P = 0.41\text{--}0.65$); that is the honest resolution limit of the deliverable — the scored comparison on one measured system.

Cross-system conclusion

The two systems above certify rather than simulate, on two real assessment records with no hydrology, no biology and no institutional history in common. Read together they support one claim about the method and one about its limits, and the second is the more useful.

What certification buys: the verdict becomes structural. The cod analysis’s central result is that the no-dominance verdict is *structural, not empirical*. Because the largest catch holding the reference point from itself is $C^* = g(\text{LRP}) - |e| = 91.59$ kt, and any rule’s protection margin is exactly C^* minus its catch there, harvest and protection are *one budget* at that point — and a reactive rule therefore cannot out-supply a flat cap it matches in protection. That is not a fact about a particular simulation run; it is a fact about two constants, and it holds for every rule in the class.

What it costs: the horizon, and where it comes from. This is the finding the two systems establish jointly and none establishes alone. **Certification has a horizon, and the horizon is set by the fitted map, not by the method.** In the cod analysis the map is expansive at the reference point for every admissible carrying capacity $K \geq 2K^*$, so the certified layer has a finite horizon: six years under the two harsher floors, seven under the informative one. In the Edwards analysis every positive-pumping certified kernel is empty beyond three years

— described there as an optimistic bound on a defect the out-of-sample audit exceeds. The two numbers differ by a factor of two and come from unrelated systems, and that is the point: the limitation is not an artefact of one specification. *A certified verdict is valid over a stated horizon, and quoting it without that horizon is quoting a different claim.* This is the practical discipline certification imposes, and it is one simulation does not impose — a simulation shows a trajectory of some length and the reader is left to supply the horizon.

What the protocol is for. Both system analyses fix the scoring criterion before any score is computed, and both report negative results. That combination is what makes a negative result informative: a rule that fails a criterion fixed in advance has been *tested*, whereas a rule that fails a criterion chosen afterwards has merely been described. The Edwards analysis is explicit that its reactive rules are retained only *nominally*, on a hybrid criterion whose wet-year margin is exactly what the robust class excludes — so the nominal retention is reported together with the reason it is nominal. A certification method that only ever produced positive verdicts would be indistinguishable from advocacy.

The design consequence, stated generally. Two results generalise beyond their systems. First, *where a threshold is protected by the weather rather than by policy, no rule in the scored family can be credited with protecting it:* The Edwards analysis finds that the 660-ft line is protected by wet years, not by the pumping family — a geometric property of scoring triggers against their own level, and one that would be invisible to any method that only asked whether the threshold was breached. Second, *a positive attractor-to-threshold margin defeats every non-negative pumping rule*, so the binding constraint is geometric rather than institutional. The cod analysis’s corresponding general statement is that harvest and protection are one budget at the reference point, so a gain in permitted catch is exactly a loss in protection margin and cannot be argued into existence.

Scope, stated once. These are scored comparisons on measured systems, and each system analysis states its own resolution limit. The Edwards analysis cannot resolve the marginal caps ($P = 0.41\text{--}0.65$) and says so; the cod analysis’s certified layer is finite-horizon by construction. The general claim is therefore not that certification resolves more than simulation — it is that *certification states what it resolves and over what horizon*, which is the property that makes a verdict usable.

The separately compiled Part II companion, *Regime Viability on the Northern Cod Stock*, uses the same cod study system but asks the distinct exact-rational, non-fitted-model question of which realized intervals certify a contraction; its arithmetic and scope are reported there, not silently attributed to the fitted-model analysis above.

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Declarations

Data availability

The analysis is reproducible: the deterministic layers are closed-form or fixed-point computations and every stochastic layer uses a fixed seed and a fixed draw budget (Section 3.8 uses $N = 20,000$ trajectories per cell, Section 3.10 $B = 2000$ bootstrap refits). All input data, analysis scripts, and result files are archived in the public repository at <https://github.com/MIKEAA2020/general-sustainability>. The primary kernel table (Table 1, Sections 3.1–3.5) is produced by `wave_e_cod/src/run_intervention_v3.py`, which writes `intervention_results_v3.json` and `intervention_boundaries_v3.csv` in `results/`; the two post-freeze reactive families of Definition 2.4 (Table 1) by `wave_e_cod/src/run_families_v3.py`, which writes `results/e2_families_v3.csv`. Re-executing either runner regenerates its output files; a verification re-execution in a fresh environment reproduced every reported value to nine significant figures; the trailing digits of the floating-point fields differ by about one part in 10^9 across BLAS builds and are not byte-identical. In the source-year convention the flat-180-kt policy’s $T = \infty$ boundary (Table 1) is empty under the 5th-percentile class, so no converged fixed point is reported for it; the runner raises the fixpoint iteration cap to 20,000 and refuses to return an unconverged iterate, which an earlier 300-iteration cap did silently on exactly this row. The critical-zone rule and cascade vocabulary follows the DFO precautionary-approach framework (DFO, 2009); the SSB series and LRP are DFO (2016) Table A2; the catch series is Schijns et al. (2021). The elevation layers of Sections 3.7–3.10 (carrying-capacity grid, stochastic viability, finite-duration floors, bootstrap bands) are produced by `wave_e_cod/src/campaign_e2_elevation_v3.py` with fixed seeds, its outputs archived in `src/results_srcyear_v3/`, and re-execution regenerates them exactly. Figures 1–9 are produced by `wave_e_cod/src/make_figs_v18.py`, which adds the two structural panels to the seven of `wave_e_cod/src/make_figs_v17.py` without changing them; v17 superseded `make_figs_v16.py` only in provenance — it imports the v3 runner and reads the v3 elevation outputs — and its output is identical. Figure 10 is produced by `wave_e_cod/src/make_figs_v19.py`; it re-derives the horizon curve from the same closed form and refuses to write the panel unless every tabulated catch reproduces the archived cadence campaign, so the figure cannot silently drift from the table. The structural constants of Section 2.4 and the corrected reactive criterion of Section 3.4 are verified cell by cell by `wave_e_cod/src/campaign_e2_structure_v3.py` (23 checks, outputs in `src/results_struct_v3/`), and the profile likelihood, joint bootstrap, observation-error sensitivity and post-moratorium windows of Sections 3.7, 3.10 and 3.12 by `wave_e_cod/src/campaign_e2_identification_v3.py` (outputs in `src/results_ident_v3/`); the horizon-in-catch invariance of Table 2 and the form-generality of Section 2.4 by `wave_e_cod/src/campaign_e2_cadence_v3.py` (outputs in `src/results_cadence_v3/`). The three Section 3.6 form rows are produced by `campaign_e2_depensation_v3.py` (free- s_0 Allee refit), `campaign_e2_allee_declared_v3.py` (declared-strength refit) and `campaign_e2_fox_form_v3.py` (Fox form), with outputs in `src/results_forms_v3/`; the Section 3.11 xteNCAM row by `arena agent 1/other documents/rerun_campaigns/campaign_e2_xteNCAM_row.py`; and the Section 3.8 one-off removal of the 1992 draw by `campaign_e2_elevation_v3.py`. The archived diagnostic `e2_breakpoint_1992.py` instead uses destination-year catch for its residuals; it is historical and supplies no source-year result claimed here. The original `campaign_e2_xteNCAM_row.py` also paired a source-year fit with destination-year floor classes. The source-year correction and a stable analytic solution of the same interior least-squares fit, archived with its original and two recomputed CSV outputs, supplies Section 3.11 instead. All other listed v3 campaigns use the declared source-year convention of Section 2.3. The registered-convention counterparts are retained rather than overwritten (`run_intervention_v2.py` and its outputs, and `wave_e_cod/src/superseded_v2/`), so the convention sensitivity listed in Section 2.3 can be

recomputed from the archive rather than taken on trust.

Edwards Aquifer. All input data, analysis scripts, and result files are archived in the public repository at <https://github.com/MIKEAA2020/general-sustainability>. J-17 daily highs: Texas Water Development Board well 6837203. Recharge: Umphres and Choi (2025), USGS data release, <https://doi.org/10.5066/P1BI62NY>. Pumpage: Edwards Aquifer Authority (2024/25), Table 1. The full intervention protocol — object, defect declaration, disturbance classes, policy family, and retention rule — was frozen and dated 2026-08-26 before any kernel, boundary, replay, or retention score was computed. It is archived with the analysis code as the frozen-protocol record, alongside the companion forecast-evaluation protocols dated 2026-08-25. The analysis is fully deterministic: re-executing the registered runner regenerates both output files, and a verification re-execution in a fresh environment reproduced both byte for byte. The machine-readable outputs include the nominal and certified retention fields and the certified-horizon record. The Section 3.6 layers (finite-duration floors; floor-class supply) are produced by `rerun_campaigns/campaign_e4_elevation.py`, archived alongside their outputs, and regenerate them exactly. The post-freeze audit layers of Sections 3.4 (closed-loop supply replay), 3.4’s current-baseline sensitivity, and 3.7 (attractor bootstrap) are produced by `wave_e_edwards/src/e4_audit_layer.py` (seeded, deterministic, on the registered panel and the paper’s declared coefficients), with outputs archived as `wave_e_edwards/results/e4_audit_layer.json`.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Code availability

The intervention record, the kernel-boundary table and every auxiliary campaign are produced by the scripts named above, archived with their outputs. The verification script `paperE2_cod_intervention_v29_verification.py` recomputes the fit, the floor classes, the kernel tables, the constructive bound, the certified horizon and the form-comparison rows from the archived campaign outputs and compares them against every value this manuscript prints; it is archived alongside this manuscript together with the mutation harness `paperE2_cod_intervention_v29_sabotage.py` and the numeric-provenance audit `paperE2_cod_intervention_v29_basis_audit.py`.

Edwards Aquifer. The kernel computation and the policy scoring are produced by the scripts named above, archived with their outputs. The verification script `paperE4_edwards_intervention_v16_verification.py` re-runs the intervention computation from the registered panel and recompares every attractor of Table 1, the smallest securing cut, the certified horizons, and the fit diagnostics against the recomputed values; it is archived alongside this manuscript.

Author contributions

A.A. conceptualized the entire work, wrote, reviewed and edited the manuscript.

AI declaration

GLM (Z.ai), Qwen (Alibaba Cloud) and DeepSeek AI assisted with drafting and iterative review. The author reviewed and edited outputs and takes responsibility for the final work.